Lesson 1 — Extra Practice

Systems of Measurement, Referents, and Measuring Instruments

Part A — Choosing a Referent That Fits the Object

A referent is useful only when it is ***close in size*** *to the object — close enough that a small number of repetitions covers it. Throughout this set use the lesson's referent values: thumb* 2.5 *cm (*1 *in), hand* 10 *cm (*4 *in), span* 20 *cm, cubit* 45 *cm, pace* 90 *cm (*1 *yd), and the height of a standard doorway,* 200 *cm.* ***Where a question states a referent's length, use that stated value****, never your own body. Round lengths to the nearest hundredth of a metre and percents to the nearest tenth of a percent unless a question says otherwise, and* ***round only at the end****.*

1. Each object below should be estimated with the referent that is *closest to it in size*. The six referents are used once each.

i. ____ length of a computer keyboard ($\approx$ 44 cm)
ii. ____ height of a refrigerator ($\approx$ 180 cm)
iii. ____ width of a postage stamp ($\approx$ 2.4 cm)
iv. ____ width of a single bed ($\approx$ 92 cm)
v. ____ length of a debit card ($\approx$ 8.5 cm)
vi. ____ width of a sheet of printer paper ($\approx$ 21.6 cm)

A. thumb width, 2.5 cm
B. hand width, 10 cm
C. span, 20 cm
D. cubit, 45 cm
E. pace, 90 cm
F. doorway height, 200 cm

(a) Write out the six matches as a list, and beside each one give the number of repetitions of the chosen referent the object needs, to the nearest whole number.

(b) Every count in (a) is about the same small number. Use that to state the rule for choosing a referent in one sentence. Then find how many thumb widths the refrigerator in (ii) would take, and say why that makes the thumb a poor choice for it.

2. A school hallway is measured with a tape as 31.50 m. Devlin estimates it with his thumb and Rue estimates it with her pace. Devlin's thumb really is 2.5 cm wide and Rue's pace really is 90 cm long, and both of them count perfectly — but each of them was given a slightly wrong *value* to multiply by.

(a) How many thumb widths does the hallway take? How many paces?

(b) Devlin multiplies his count by 2.4 cm. State his estimate in metres, the error in cm and in m, and the error as a percent of 31.50 m.

(c) Rue multiplies her count by 88 cm. Answer all of (b) again for Rue.

(d) Rue's referent value is wrong by 2 cm — twenty times Devlin's 0.1 cm — and yet her estimate is much the better one. Explain why, and write the rule the comparison illustrates as a single sentence beginning "The error in an estimate is the number of repetitions times…".

3. Jamil is asked to estimate the width of a smartphone, which is 7.0 cm, and he proposes to use his pace (90 cm).

(a) How many paces wide is the phone? Explain why the number you obtain is not something a person can judge by eye.

(b) Suppose Jamil can judge a length only to the **nearest half-pace**. The largest error that leaves is half of a half-pace. Find that error in cm, and express it as a percent of 7.0 cm.

(c) Still reading to the nearest half-pace, find the **shortest** object whose length a pace can estimate to within 5%.

(d) Question 2 showed what goes wrong when a referent is far too small, and this question shows what goes wrong when it is far too large. State the two-sided rule in one sentence, naming the quantity that blows up at each end.

Part B — The Estimate Is Count × Referent Length, Read Both Ways

One relation carries this whole lesson: estimate = count × referent length. Questions 4–6 run it forwards, then solve it for the count, then solve it for the referent length.

4. Find each estimate, in centimetres and then in metres.

(a) 14 hand widths, hand = 10 cm

(b) 23 paces, pace = 90 cm

(c) 6 cubits, cubit = 45 cm

(d) 38 thumb widths, thumb = 2.5 cm

5. Now solve for the count. In each case state how many whole repetitions of the referent the estimate represents.

(a) an estimate of 7.20 m using a span of 20 cm

(b) an estimate of 54.00 m using a pace of 90 cm

(c) an estimate of 2.60 m using a cubit of 45 cm

(d) Part (c) does not come out whole. Round the count to the nearest whole cubit, state the estimate that whole count produces, and find the percent error that the rounding alone introduces. Then name a referent from the list that would have made the rounding matter less, and say why.

6. Priya paces a corridor that a tape measure gives as 24.00 m, and it takes her exactly 25 paces.

(a) Solve estimate = count × referent length for the referent length, and find Priya's true pace in centimetres.

(b) Priya has always used 90 cm as her pace. Were her past estimates too large or too small? By what percent of the true length?

(c) Last week Priya reported a fence as 18.00 m, using 90 cm per pace. Recover her pace count, then give the fence's true length to the nearest hundredth of a metre.

Part C — Accumulated Error: What the Count Multiplies

A referent whose value is wrong is wrong every single time it is laid down. *Questions 7–9 measure that, and then separate the part of the error that grows with the count from the part that does not.*

7. Theo's span is truly 19.5 cm, but he uses 20 cm as its value.

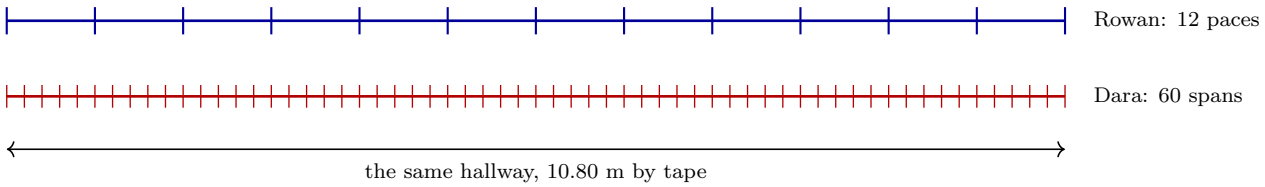
(a) He lays his span end to end 20 times along a countertop. Give the countertop's true length and Theo's estimate, both in metres, and the error in centimetres. Show that the error is 0.5 cm times the count.

(b) He then measures a wall the same way and counts 60 spans. Give the true length, the estimate and the error again.

(c) Find the percent error of each of the two estimates, to the nearest tenth of a percent.

(d) The error in centimetres tripled from (a) to (b), but the percent error did not move at all. Explain why, and say which of the two numbers a person should quote when reporting how good an estimate is.

8. Dara and Rowan estimate the same hallway, which a tape gives as 10.80 m. Dara's span is truly 18.0 cm and she uses 20 cm as its value. Rowan's pace is truly 90.0 cm and he uses 95 cm as its value. Both count perfectly.



(a) Confirm from the true referent lengths that the counts in the diagram are right, then give each student's estimate in metres.

(b) Find each student's percent error. Then show that each one equals the error in that student's *referent value* expressed as a percent of the true referent length.

(c) Rowan's referent value is wrong by 5 cm and Dara's by only 2 cm, yet Rowan's estimate is twice as good. Explain the apparent contradiction in one or two sentences.

(d) Suppose instead that both students knew their referent lengths exactly, but each placement can still be out by up to 0.4 cm through imperfect alignment. Find the worst-case total for each student, and state what that adds to the case for the larger referent.

9. An object of true length L is covered by exactly n repetitions of a referent whose true length is t, so that $L = nt$. The estimate is made by multiplying the count by an assumed referent length a.

(a) Write the estimate E in terms of n and a.

(b) Write the absolute error $E - L$ in terms of n, a and t, factored.

(c) Write the percent error and simplify it as far as it will go.

(d) Your answers to (b) and (c) say two different things about using a small referent many times. State each in one sentence, and check both against the numbers in question 7.

(e) Name the one thing a real estimate suffers from that part (c) cannot see, say whether it grows with n, and name the earlier question that measures it.

Part D — Percent Error, and "Good Enough for What?"

For an estimate compared with a measured value,

$$\textit{percent error} = \frac{|\,\textit{estimate} - \textit{measured}\,|}{\textit{measured}} \times 100\%.$$

An estimate is never good or bad on its own. It is good enough, or not good enough, ***for a stated purpose****.*

10. Find the percent error of each estimate, to the nearest tenth of a percent.
(a) estimate 3.40 m, measured 3.25 m (b) estimate 78 cm, measured 82 cm (c) estimate 12.0 m, measured 12.6 m

(d) Rank the three from best to worst. One of them has by far the *smallest* error in centimetres and yet is the worst of the three. Identify it, and say what that shows about judging an estimate by its absolute error.

11. A rectangular room's two walls are measured with a tape as 2.48 m and 3.10 m. Estimating with a pace, a student reports them as 2.70 m and 3.60 m.

(a) Find the percent error of each of the two estimates.

(b) **Hanging a shelf.** A shelf is to be cut to fit snugly between the two walls that are 2.48 m apart. Decide whether the estimate is good enough, and support the decision with the number of centimetres by which the shelf would be wrong.

(c) **Ordering flooring.** Laminate is sold only in whole boxes, each covering 2.2 m^2, and the installer adds 10% for waste before rounding up to the next whole box. Work out the number of boxes from the measured lengths and then from the estimated lengths, and state what the estimate costs.

(d) One of these two purposes tolerates a 9% error and the other does not. Say which is which, and describe in general the feature that makes a purpose tolerant of estimation error.

Part E — Referents in Both Systems, and Crossing Between Them

Use the conversions the unit uses: $1\ in = 2.54\ cm$, $1\ ft = 0.3048\ m$, $1\ m = 3.2808\ ft$, $1\ ft = 12\ in$, $1\ yd = 3\ ft$. *Two estimates can only be compared after they are expressed in the* ***same*** *unit.*

12. Ori estimates the length of a desk twice: once with his span, which he takes to be 20 cm, and once with the length of his shoe, which he takes to be 12 in. He counts 9 spans, and separately 6 shoe lengths.

(a) Give the SI estimate in centimetres and in metres.

(b) Give the imperial estimate in inches, then in feet, then in yards.

(c) Convert the imperial estimate to centimetres, and state by how many centimetres the two estimates disagree and what percent of the imperial estimate that is.

(d) Both estimates came from the same desk and the same person. Give the most likely reason they disagree, and name the single measurement that would settle which of the two referent values is at fault.

13. Two garden beds are estimated by two different people. Bed A is paced out at 5 paces, with the pace taken as 90 cm. Bed B is stepped out at 16 of a friend's shoe lengths, with the shoe taken as 1 ft.

(a) Express both estimates in metres, to the nearest hundredth of a metre.

(b) Now express both in feet instead, to the nearest tenth of a foot, and check that the two beds come out in the same order either way.

(c) State the difference between the two beds in metres, and as a percent of the longer estimate.

(d) Each estimate could be out by as much as 5%. Work out the interval of possible true lengths for each bed, and then decide honestly whether the evidence shows that bed B is longer than bed A.

Part F — When a Referent Will Not Do

Estimation is the right tool when the decision has a margin built into it, and the wrong tool when a piece is cut, machined or set to the number itself.

14. For each task, state whether a personal referent is good enough or a measuring instrument is required. If an instrument is required, name one and the precision it reads to; in every case give the **tolerance** — how far out you may be before the task fails — that decides the matter.

(a) deciding whether a 1.90 m sofa will fit along a wall you have paced at about 3 m

(b) cutting a replacement pane of glass to sit in an existing window frame

(c) judging whether a trail is closer to 5 km or to 10 km

(d) turning a steel bolt to fit a hole of diameter 12.00 mm

(e) setting a basketball hoop to the regulation height of 3.05 m

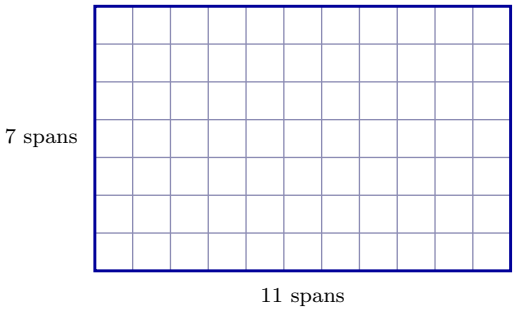
(f) pegging out a vegetable bed that is to be "about 3 m by 5 m"

(g) Sort your six answers into the two groups and complete this sentence: "A referent is good enough exactly when…".

Part G — Estimating a Perimeter and an Area

A referent estimates a length. *A perimeter is a sum of lengths and an area is a product of them — and a product treats error very differently from a sum.*

15. A rectangular rug is covered by Mira's span, taken as 20 cm: it is 11 spans long and 7 spans wide, with no gaps and no overlaps.



(a) Give the rug's estimated length and width in metres, then its estimated area in square metres.

(b) Mira's span is really 19 cm. Give the rug's true length, width and area.

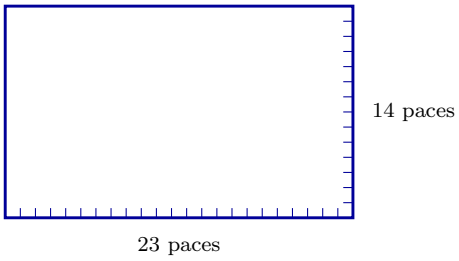
(c) Find the percent error in each *length*, and then the percent error in the *area*, to the nearest tenth of a percent.

(d) The area's percent error is a little more than double the length's. Show exactly where the factor comes from by writing the ratio of estimated area to true area as the square of a single fraction, and account for the "a little more".

(e) Predict, without computing an area, the percent error you would get in the *volume* of a box whose three dimensions were all estimated with the same span. State the general rule for a measurement in k dimensions.

16. A rectangular vegetable plot is paced out as 23 paces by 14 paces, with the pace taken as 90 cm.

(a) Give the perimeter first as a count of paces, then in metres.



(b) Fencing is sold in rigid panels 2.5 m long and cannot be cut. How many panels does the estimate call for?

(c) The gardener's pace is really 87 cm. Find the true perimeter, the number of panels actually needed, and the percent error of the estimate.

(d) The percent error in (c) is small, and yet the estimate still orders the wrong number of panels. Explain how a small percent error can produce a whole extra panel, and say what the gardener should do before ordering.

Part H — Error Analysis: Three Students, Three Different Failures

Each student below made exactly one error. Find it, name it in words, give the corrected work, and state the rule that would have prevented it.

17. Asked to estimate the length of a running track's straight, Jon used his hand width (10 cm). He counted the hands three separate times and got 850, then 910, then 870. He reported the first count: "about 85 m".

(a) Convert all three counts to metres. Jon's arithmetic is correct every time, so name the error anyway — and be precise about which part of count × referent has gone wrong.

(b) Find the spread of his three answers in metres, and express that spread as a percent of their mean.

(c) The straight is later measured at 87.50 m. Find the percent error of each of Jon's three answers, and say what the pattern of the three tells you about the direction of his error.

(d) Redo the estimate with a pace of 90 cm: how many paces is 87.50 m, to the nearest whole pace? Give two separate reasons the pace is the better referent here.

18. Asked to estimate the width of a classroom, Amara stepped it out in 24 shoe lengths, took her shoe to be 12 in, and converted:

$$24 \times 12 = 288 \text{ in}, \qquad 288 \times 2.54 = 731.52 \text{ cm}, \qquad \text{width} = 7.315 \text{ m}$$

(a) Check her conversion arithmetic. Is a single keystroke wrong?

(b) Name her error, and say where the three decimal places actually came from.

(c) Amara's shoe is somewhere between 29 cm and 32 cm long. Work out the interval her estimate really lies in, and state its width in centimetres.

(d) Rewrite her answer as it should have been reported, and state the general rule about the precision of an estimate in a single sentence.

19. Estimating the length of a hallway, Kofi paced as far as a pillar and counted 20 paces (90 cm each). Past the pillar the floor was cluttered, so he switched to spans (20 cm) and counted 16 of them. He reported

$$(20 + 16) \times 90 = 3\,240 \text{ cm} = 32.40 \text{ m}$$

(a) Name the error in words, saying exactly what Kofi added together that cannot be added.

(b) Give the correct estimate in centimetres and in metres, and write the general formula for an estimate that switches referents once.

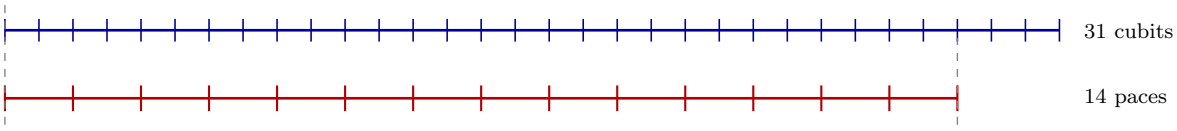
(c) Find the size of Kofi's error in metres, and as a percent of the correct estimate.

(d) Rework the second stretch as a number of *paces* instead, then finish the estimate with a single multiplication. Confirm that it agrees with (b), and say why this route is the safer habit.

Part I — Stretch: Two Students, Two Referents, One Wall

When two honest estimates disagree, the disagreement is itself information: it tells you how far apart the two referent values must be, and it tells you what to go and measure.

20. Nadia and Beckett each estimate the horizontal run of a climbing wall. Nadia counts 31 cubits and takes her cubit to be 45 cm. Beckett counts 14 paces and takes his pace to be 90 cm.



drawn to scale: 1 cm on the page = 1 m of wall

(a) Give both estimates in metres, the difference between them, and that difference as a percent of Beckett's estimate.

(b) Suppose Nadia's estimate is the correct one. What would Beckett's true pace have to be? Now suppose Beckett's is correct: what would Nadia's true cubit have to be? Give both to the nearest tenth of a centimetre.

(c) Neither of the two referent lengths in (b) is impossible for a real person. Say so with a reason for each, and explain why that means the disagreement cannot be settled by calling one estimate absurd.

(d) Decide which estimate is *likelier* to be the closer one. Give one argument from the mathematics of Part C and one argument against your own choice, then state your verdict.

(e) Name the extra information that would settle the disagreement, and distinguish between the measurement that settles *how long the wall is* and the measurement that settles *whose method was better*.

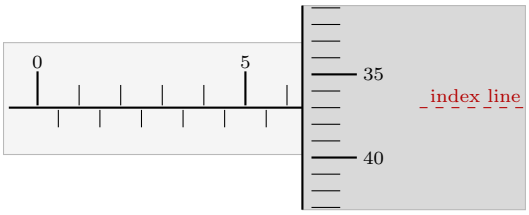
(f) A tape now gives the wall as 13.10 m. Find each student's percent error, then find each student's true referent length, to the nearest tenth of a centimetre. Say whether the verdict you gave in (d) survives.

Part J — Reading a Vernier Caliper and a Micrometer (Section 5)

Every precision instrument carries a fixed scale that gives the ***coarse reading*** *and a moving scale that gives the* ***fine reading****; the measurement is the sum of the two. The diagrams in this set are drawn as the instruments really are. On a caliper the vernier's own* **0** *mark is the index: the coarse reading is the last whole millimetre that mark has* passed*, and the fine reading is* 0.05 *mm times the number of the vernier division that lines up with a main-scale mark. On a micrometer the coarse reading is the last sleeve mark the thimble's edge has* uncovered *— and the sleeve is marked every* 0.5 *mm, with the half-millimetre marks below the index line — while the fine reading is* 0.01 *mm times the thimble division standing at the index line. Name both parts every time, then add them. Caliper readings are stated to* 0.05 *mm (two decimals), micrometer readings to* 0.01 *mm. Use exact* π*, never* 3.14*, and round only at the very end.*

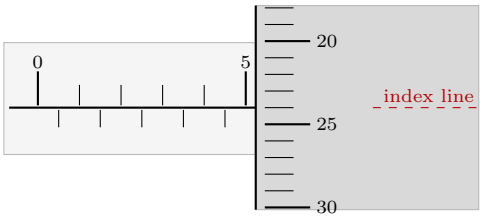
21. For each micrometer below, state the coarse reading and the fine reading separately (show the multiplication for the fine reading), then add them. Finish by stating how far the thimble must still turn — in complete revolutions and divisions — to bring the reading up to the next whole millimetre.

(a)



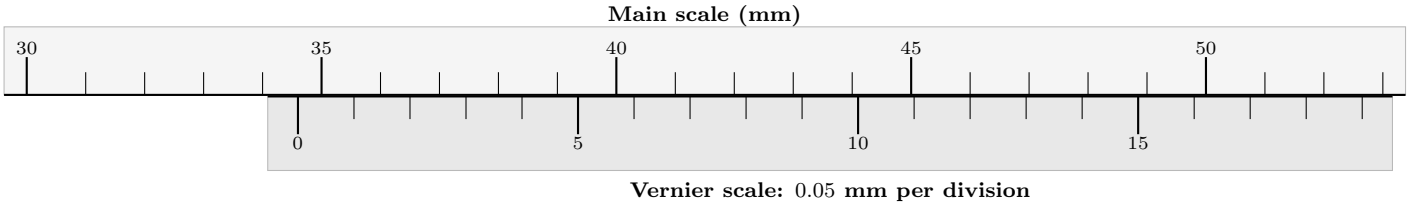
Sleeve: a mark every 0.5 **mm**
Thimble: 0.01 **mm per division**

(b)



Sleeve: a mark every 0.5 **mm**
Thimble: 0.01 **mm per division**

22. A vernier caliper reads to 0.05 mm.

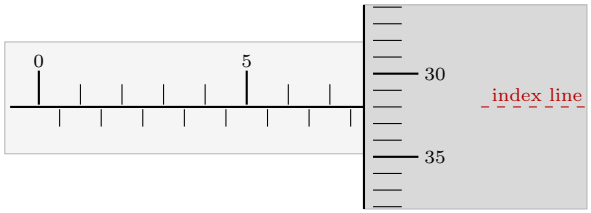


Vernier scale: 0.05 **mm per division**

(a) State the coarse reading, the fine reading (with the multiplication), and the total, written to the caliper's own precision.

(b) How much further would the slide have to move for the vernier's 0 to reach the next whole millimetre? Which vernier division would line up then, and what would the reading be?

23. **The half-millimetre line.** A micrometer reads as shown.



Sleeve: a mark every 0.5 **mm** **Thimble:** 0.01 **mm per division**

(a) State the coarse reading, the fine reading, and the total.

(b) Was the coarse reading given by a whole-millimetre mark or by a half-millimetre mark? Say how the diagram tells you.

(c) A student takes the coarse reading from the last *whole*-millimetre mark instead. What do they report, and by how much are they out? Explain why this particular mistake is always the same size.

24. Name the best instrument for each job — trundle wheel, tape measure, ruler, vernier caliper, or micrometer — and give a reason that mentions *both* the size of the thing being measured and the precision the job needs.

(a) The diameter of a copper wire, for a current-carrying calculation.

(b) The inside diameter of a pipe fitting, about 30 mm, needed to 0.1 mm.

(c) The perimeter of a school field.

(d) The length and width of a classroom, for a flooring order.

(e) The thickness of one sheet of paper.

(f) The length of a bolt, about 60 mm, needed to the nearest millimetre.

Lesson 2 — Extra Practice

Converting Units Within and Between SI and Imperial

Part A — The Unit Factor: Writing a Conversion So the Unwanted Unit Cancels

Every conversion in this set must be written as ***unit analysis****: the given measurement multiplied by one or more* ***unit factors****, each a fraction built from a known equality and turned so that the unwanted unit sits* opposite *where it started and cancels. From* $1\ m = 100\ cm$ *come the two factors* $\frac{100\ cm}{1\ m}$ *and* $\frac{1\ m}{100\ cm}$*; they are both equal to* 1*, and choosing between them is the entire skill.* ***Carry full precision through every step and round only once, at the end. Every answer carries its unit.*** *This set is* ***linear units only****.*

1. Convert 4.28 km to millimetres using unit analysis, in two factors, through metres.

(a) From 1 km = 1000 m, write **both** fractions that can be built. State which one converts kilometres *into* metres, and justify your choice by naming the unit that cancels — not by saying which answer "looks right".

(b) Write the full two-factor chain and evaluate it. Show the cancellations. Give the answer in standard notation **and** in scientific notation.

(c) A classmate started with $\frac{4.28\text{ km}}{1} \times \frac{1\text{ km}}{1000\text{ m}}$. Nothing cancels. State the unit this product actually carries, and explain why a chain that leaves a nonsense unit is a warning that fires **before** any arithmetic is done.

2. Convert each measurement. Each crosses **more than one** prefix, so each needs **at least two** unit factors — route every one through the base unit (the metre). Show the factors and the cancellation, not just the answer.

(a) 3400 mm = _____ km

(b) 620 μm = _____ cm

(c) 0.45 Gm = _____ km

(d) 92 cm = _____ nm

3. Each conversion below crosses so many powers of ten that standard notation is unreadable. Convert using unit factors and leave every answer in **scientific notation**, $a \times 10^n$ with $1 \leq a < 10$.

(a) 5.6 nm = _____ km

(b) 7.2 Mm = _____ mm

(c) 148 μm = _____ km

(d) Arrange these three lengths from **shortest to longest**: 4.2×10^5 μm, 4.1×10^{-4} km, 415 mm. Convert all three to a **common unit** first, and say why comparing the leading digits 4.2, 4.1 and 415 as they stand proves nothing at all.

Part B — Inside the Imperial System: Four Different Multipliers

The metric prefixes are all powers of ten. The imperial units are not: 12 *inches to a foot,* 3 *feet to a yard,* 5280 *feet or* 1760 *yards to a mile. Unit factors handle them without any change of method — which is the point. Name the fact each factor comes from.*

4. Convert each of the following using unit factors. Beside each factor, write the conversion fact it came from.

(a) 7 yd = _____ in (two factors)

(b) 0.35 mi = _____ ft

(c) 4631 in = _____ ft _____ in

(d) 2.4 mi = _____ yd

5. A surveyed fence line measures 1875 inches.

(a) Express the length in **feet and inches**. Show the division and name the remainder's unit.

(b) Now express it in **yards, feet and inches**, continuing from your answer to (a) rather than starting over.

(c) Express the same length as a **decimal number of yards**, to four decimal places, using a single unit-factor chain.

(d) Your answers to (b) and (c) describe the same fence. Explain, in one sentence, why the digits after the decimal point in (c) are **not** the "feet and inches" part of (b).

6. A carpenter cuts three shelves of lengths 4 ft 9 in, 3 ft 8 in and 5 ft 11 in from a single 16-foot board. Ignore the width of the saw cut.

(a) Convert all three lengths to a **single** unit, add them, and report the total in feet and inches. Say why converting to one unit first is safer than adding "feet to feet and inches to inches".

(b) Is the 16-foot board long enough? Show the comparison in a single unit, and give the offcut left over in feet and inches.

Part C — Crossing Between the Systems

Use only the booklet's factors: 1 *ft* = 0.3048 *m and* 1 *ft* = 30.48 *cm. A measurement written in two imperial units must be collapsed into* ***one*** *imperial unit before the crossing factor is applied.*

7. Convert each measurement, using unit factors throughout and rounding only at the end to the precision stated.

(a) 23 ft = _____ m (nearest thousandth of a metre)

(b) 6 ft 4 in = _____ cm (nearest tenth of a centimetre)

(c) 2.15 m = _____ ft _____ in (nearest inch)

(d) 85 cm = _____ in (nearest tenth of an inch)

8. A doorway is measured as 205 cm tall. Two crossing factors are available: 1 ft = 0.3048 m and 1 ft = 30.48 cm.

(a) **Choose** one of the two factors and justify the choice by counting the unit factors each route needs. Then convert the height to feet, carrying full precision.

(b) Redo the conversion with the **other** factor and confirm the two agree. Explain what makes them agree — state the exact relationship between 0.3048 and 30.48.

(c) Report the doorway height in **feet and inches**, to the nearest inch, and state which of your two routes you would teach to someone meeting this for the first time, and why.

9. A conversion fact you were never given can always be **built** out of two you were. Use only 1 ft = 0.3048 m, 1 yd = 3 ft, 1 mi = 5280 ft and 1 km = 1000 m.

(a) Build 1 yd = _____ m as an **exact** value. Show the chain.

(b) Build 1 mi = _____ km as an **exact** value. Show the chain.

(c) The booklet's table lists 1 mi = 1.6093 km. Compare your exact answer from (b) with that table value: state the difference per mile, and say how far apart the two factors would place the finish line of a 26.2-mile marathon. Answer in metres, to the nearest tenth.

Part D — Rates: Two Units Change, and One of Them Is Underneath

*A rate such as km/h is a fraction. A unit in the **denominator** is cancelled by a factor carrying that unit in its **numerator** — upside down compared with everything in Parts A to C. Say out loud which unit you are cancelling and where it sits, every time.*

10. A highway speed is posted as 126 km/h.

(a) Convert it to metres per second. Write the chain in full, and beside each factor state **which unit it cancels and whether that unit was on the top or the bottom** of the rate.

(b) Convert 14 m/s back to kilometres per hour. State which two factors are now inverted compared with (a), and why.

(c) Without converting anything, explain why the m/s figure must be the *smaller* number of the two in part (a). Refer to both the numerator unit and the denominator unit in your answer.

11. Trim moulding is advertised at \$3.45 per foot at Store A, and at \$10.95 per metre at Store B.

(a) Convert Store A's price to dollars per metre, to the nearest cent. The unit **foot** sits in the denominator — state which way up the unit factor must therefore be written, and show the cancellation.

(b) Which store is cheaper, and by how much per metre? Answer in a sentence, with units, to the nearest cent.

(c) Before calculating, a student predicted that the price per metre would be *larger* than the price per foot. Was the prediction right, and what one fact about the two units makes it certain without any arithmetic?

12. A conveyor belt at a gravel pit moves at 45 feet per minute.

(a) Convert the speed to kilometres per hour, to the nearest hundredth. Three unit factors are needed; list them before you start and say what each one is for.

(b) The belt runs for a 7.5-hour shift. How far does a single stone on the belt travel, in kilometres, to the nearest tenth? Use the **unrounded** rate.

Part E — Comparisons and a Decision You Cannot Make in Fractions

*A comparison between systems is not answerable until both quantities are in one unit. Each answer here is a **sentence**, with the difference stated and carrying its unit.*

13. Which is longer, 5 ft 8 in or 1.75 m?

(a) Convert 5 ft 8 in to metres, carrying full precision. Collapse the mixed measure to a single imperial unit first.

(b) Answer the question in a **sentence**, stating which is longer and by how much, in centimetres, to the nearest tenth.

14. One trailhead sign gives the loop as 8.4 km; an older sign on the same loop gives it as 5.2 mi. Use the booklet's factor 1 mi = 1.6093 km.

(a) Convert the older sign's distance to kilometres and state, in a sentence, which sign claims the longer loop and by how many **metres**, to the nearest tenth.

(b) The difference is small. Give one reason the two signs might disagree that has nothing to do with anyone making a conversion error.

15. A garden plan drawn in metres calls for 23.4 m of wooden edging in total. The lumberyard sells that edging **only** in 12-foot boards.

(a) Convert one board to metres exactly, then find how many boards the plan needs. Give the **unrounded** quotient first, then the number of boards to buy, and **state in words why that number is rounded the way it is**.

(b) How much edging is left over, in metres to the nearest hundredth? Give it also as a percentage of one board, to the nearest whole percent.

(c) Part (a) assumed offcuts may be joined end to end. If instead each straight run of the border had to be one uncut piece, would your answer to (a) still be a lower bound, an upper bound, or neither? Justify.

Part F — Error Analysis: Three Students, Three Different Failures

Each student below made exactly one error. Name it in words, give the corrected unit-factor work, and state the ***reasonableness check*** *— a sentence about the relative size of the two units — that would have caught it before any marking.*

16. Asked to convert 4500 mm to kilometres, a student wrote

$$\frac{4500\text{ mm}}{1}\times\frac{1000\text{ mm}}{1\text{ m}}\times\frac{1000\text{ m}}{1\text{ km}}=4\,500\,000\,000\text{ km}$$

(a) Both factors are upside down. Show what happens to the units when you try to cancel, and state the unit the student's product actually carries.

(b) Give the corrected chain and the correct answer, in scientific notation and in standard notation.

(c) The student's answer, 4.5×10^{9} km, is about the distance from the Sun to Neptune. Write the one-sentence reasonableness check — comparing the *size* of a millimetre with the size of a kilometre — that rules the answer out instantly, and say why that check is worth more than re-reading the arithmetic.

17. Asked to convert 0.47 m to millimetres, a student reasoned: "metres to millimetres is three places on the number line," moved the decimal three places, and wrote 0.00047 mm.

(a) The student's **count** of three places was right. Name precisely what was wrong, and give the correct answer.

(b) Write the conversion as a unit factor. Explain how that notation makes the direction impossible to get wrong, whereas counting places on a number line leaves it to be remembered.

(c) State the reasonableness check for this one, and give the ratio between the student's answer and the correct answer.

18. Asked to convert 5 ft 8 in to metres, a student wrote 5 ft 8 in = 5.8 ft, then $5.8 \times 0.3048 = 1.76784$, and reported 1.768 m.

(a) The second step is flawless. Name the error in the first step, and state what length "5.8 ft" actually is, in feet and inches.

(b) Give the correct conversion of 5 ft 8 in to metres, to the nearest thousandth, and the size of the student's error in centimetres, to the nearest tenth.

(c) Question 13 asked which is longer, 5 ft 8 in or 1.75 m. State what conclusion this student would have reached there, and what that shows about errors too small to look suspicious.

Part G — Stretch: One Chain, Two Systems, and a Rate

Every conversion in this set — metric, imperial, crossing, and rate — is the same move written the same way. Question 19 uses all four at once, in a single line.

19. A crew lays pipeline at a steady 385 feet per hour. The pipeline is 12.5 km long.

(a) Using only 1 ft = 0.3048 m and 1 km = 1000 m, build the factor 1 km = _____ ft, to four decimal places. Show the chain.

(b) Convert the crew's rate to kilometres per hour. Give the **exact** value, not a rounded one, and keep it for part (c).

(c) How many hours will the job take, to the nearest tenth? Use the unrounded rate from (b). Then express the time in **days and hours**, to the nearest hour.

(d) Now write the whole problem as **one** unit-factor chain, starting from 12.5 km and ending in hours, with no intermediate value written down at all. Confirm it gives the same answer.

(e) In the single chain of part (d), the rate 385 ft/h appears as a unit factor. Write it out and say which way up it had to be written and why — then state the general rule your answer illustrates about using a *rate* as a conversion factor.

Lesson 3 — Extra Practice

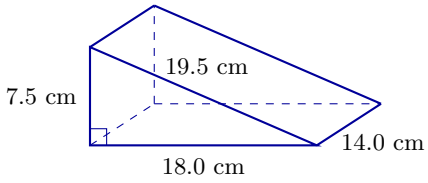
Surface Area and Volume of Prisms and Cylinders

Part A — One Formula, Three Bases: $V = A_{\text{Base}} \times h$

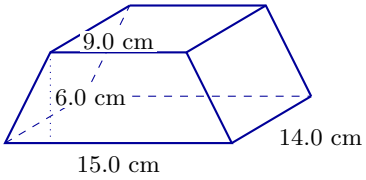
*Every right prism and every right cylinder obeys the same rule: $V = A_{Base} \times h$. Only the **base area** changes from solid to solid. Throughout this set use **exact** π — never 3.14 — carry full precision through every intermediate step, and **round only once, at the very end**, to the precision each question names. **Every answer carries its unit**: cm^2 for area, cm^3 for volume.*

1. The three solids below all have the same height, $h = 14.0$ cm. Only the base changes. For each solid, find (i) the area of one base and (ii) the volume, to the nearest tenth.

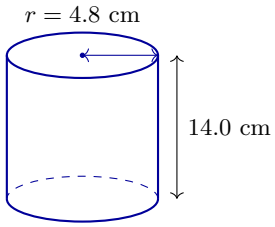
(a) right triangular prism



(b) right trapezoidal prism

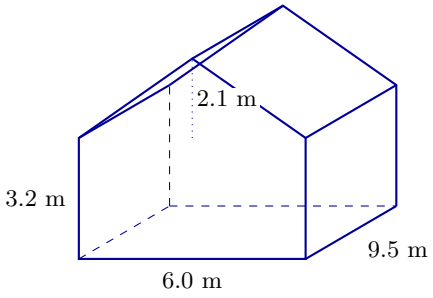


(c) right cylinder



(d) Write down the one formula you used all three times. Then name the *only* quantity that changed from (a) to (c), and explain in one sentence why a cylinder is handled by the same rule as a prism even though its base has no straight sides.

2. A storage shed is a prism whose base is the cross section shown — a rectangle with a triangular roof on top. The shed is 9.5 m long.



(a) Find the area of the cross section, in m^2.

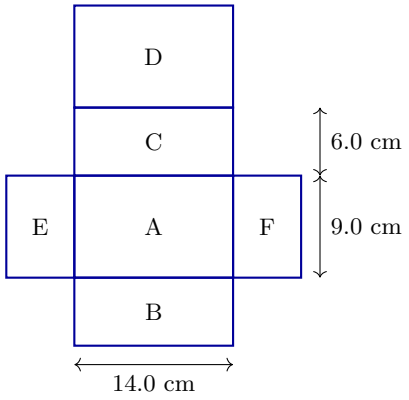
(b) Find the volume of the shed, in m^3.

(c) Express the volume of the shed in litres. *(Recall from Section 3 that 1 $m^3 = 1000$ L.)*

Part B — Surface Area Read Off a Net

*A **net** is the solid unfolded flat. Nothing is hidden in a net, so a surface area found from one is just a sum of areas of flat shapes — provided you count every face exactly once.*

3. The net below folds into a right rectangular prism.



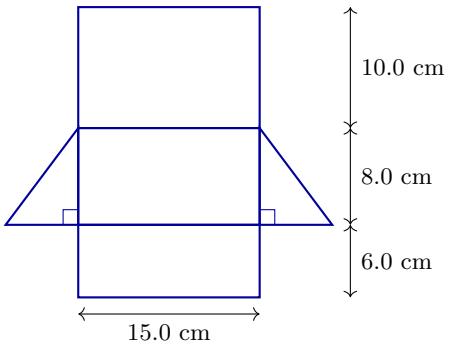
(a) Copy and complete the table: for each of the six faces A–F, give its two dimensions and its area. Then state which faces are congruent to which.

Face	Dimensions	Area
A	_____	_____
B	_____	_____
C	_____	_____
D	_____	_____
E	_____	_____
F	_____	_____

(b) Total the six areas to get the surface area of the prism.

(c) Find the volume of the prism. Explain why the net tells you the surface area immediately but does *not* give you the volume without further thought.

4. The net below folds into a right triangular prism whose triangular ends are right triangles.

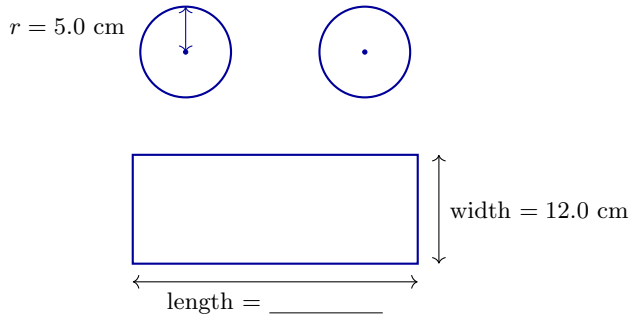


(a) There are five faces. Name them (two of one kind, three of another) and give the area of each.

(b) Find the surface area of the prism.

(c) Show that the three rectangles together have area equal to (perimeter of the triangle) $\times$ 15.0 cm, and say why that shortcut always works for a right prism.

5. The net of a right cylinder is two circles and one rectangle. The cylinder below has radius 5.0 cm and height 12.0 cm.

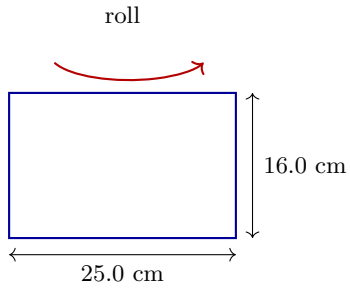


(a) The rectangle's *length* is the blank in the diagram. Say what measurement of the cylinder it equals, and why. Then compute it to the nearest hundredth of a centimetre.

(b) Find the area of the rectangle and the combined area of the two circles, each to the nearest tenth. Add them to get the surface area of the cylinder.

(c) Find the volume of the cylinder to the nearest tenth, and state the one number in this question that is used for the volume but plays no part in the surface area of the two circles.

6. A flat rectangular sheet of tin measures 25.0 cm by 16.0 cm. It is rolled into an open tube (no ends) so that two opposite edges meet exactly, with no overlap.



(a) Suppose the 25.0 cm edges are the ones brought together, so the 16.0 cm side becomes the circumference. Find the radius of the tube to the nearest hundredth of a centimetre, and then its volume to the nearest tenth.

(b) Now roll the same sheet the other way, so the 25.0 cm side becomes the circumference. Find the new radius and the new volume, to the same precision.

(c) The two tubes are made from exactly the same amount of tin, so their curved surface areas are equal. Explain why their volumes are not, and state which way of rolling gives the larger volume.

Part C — Running the Formulas Backwards

When a volume or a surface area is the given *and a dimension is the unknown, substitute everything you know into the formula and solve the equation. A radius hiding inside r^2 needs a square root; an edge hiding inside s^2 does too.*

7. Solve each of the following for the dimension named. Show the substitution and the rearrangement, and round only at the end.

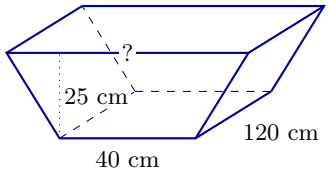
(a) A right prism has volume 1890 cm^3 and a base of area 63.0 cm^2. Find its height.

(b) A right cylinder has volume 2400 cm^3 and height 15.0 cm. Find its radius, to the nearest tenth of a centimetre.

(c) A cube has surface area 384 cm^2. Find the length of one edge, then the volume of the cube.

(d) In exactly one of (a), (b), (c) the unknown could be found by a single division. Say which, and say what it is about the *other two* formulas that makes a division alone insufficient.

8. A water trough is a right prism whose base is a trapezoid. The trough is 120 cm long and 25 cm deep, the bottom of the trapezoid is 40 cm across, and the trough holds exactly 165 L when filled to the brim.



(a) Convert the capacity to cm^3, then find the area of the trapezoidal base.

(b) Use the trapezoid area formula $A = \frac{1}{2}(a + b)h$ to find the width across the top of the trough. State the unit.

9. The **curved** surface of a cylindrical hot-water tank — the side only, with neither circular end — has area 1508.0 cm^2. The tank is 24.0 cm tall.

(a) Write the formula for the curved surface area alone, then solve it for the radius, to the nearest tenth of a centimetre.

(b) Using your radius from (a), find the *total* surface area of the closed tank, to the nearest tenth.

(c) Find the volume of the tank, to the nearest tenth, and state its capacity in litres.

Part D — Composite Solids: Surfaces Are Added *and* Removed

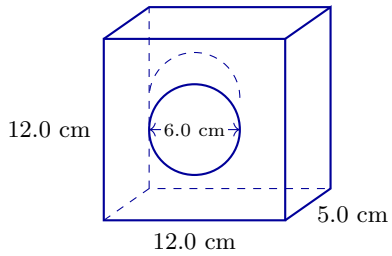
Read this before you start. When two solids are joined or one is cut out of another, the volumes simply add or subtract — but the ***surface area does not****. Two things go wrong most often:*

- *the faces that are now buried, or drilled away, must be* ***subtracted****;*
- *any* ***new*** *surface exposed by the cut — the inside wall of a hole — must be* ***added****.*

Write the removals and the additions as a separate line of your work every time.

10. A rectangular steel plate measures 12.0 cm by 12.0 cm and is 5.0 cm thick. A circular hole of diameter 6.0 cm is drilled straight through the 5.0 cm thickness.

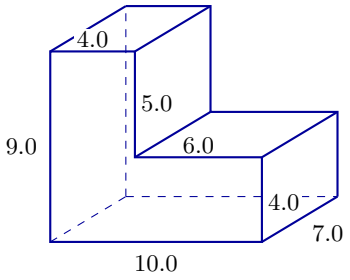
(a) Find the volume of the plate that remains, to the nearest tenth.



(b) Find the surface area of the plate that remains, to the nearest tenth. Set your work out in three separate lines: the six faces of the solid plate, then the area **removed**, then the area **added**.

(c) A classmate says "drilling a hole takes metal away, so the surface area must go down." Say whether the surface area went up or down here, by how much, and explain in one sentence why removing material can *increase* surface area.

11. The solid below is a right prism whose base is an L-shape. All dimensions are in centimetres.



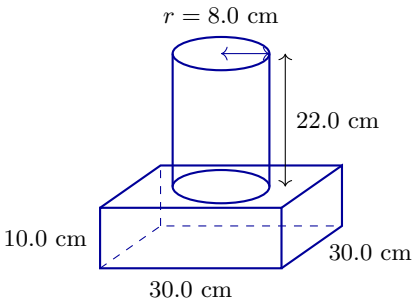
(a) Find the area of the L-shaped base, by splitting it into two rectangles *and* by subtracting a rectangle from a larger one. The two methods must agree.

(b) Find the volume of the solid.

(c) Find the perimeter of the L-shaped base, then the total area of all the rectangular side faces, then the surface area of the whole solid.

(d) Compare the perimeter you found in (c) with the perimeter of the 10.0 cm by 9.0 cm rectangle that the L fits inside. They are equal. Explain why cutting the notch out of the corner changed the *area* of the base but not its *perimeter*.

12. A concrete garden ornament is a cylinder sitting centred on top of a square prism. The prism is 30.0 cm by 30.0 cm by 10.0 cm; the cylinder has radius 8.0 cm and height 22.0 cm. The ornament rests on the ground, and every surface *except* the bottom of the prism is to be sealed.



(a) Find the total volume of the ornament, to the nearest tenth.

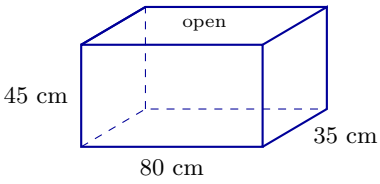
(b) Find the total area to be sealed, to the nearest tenth. List every surface you are counting, and every surface you are leaving out, before you add anything up.

(c) Two circles of the same size appear in your work in (b): one is subtracted and one is added. Name them, explain where each comes from, and say what their combined effect on the answer is.

Part E — Open Containers: the Wording Decides the Faces

Nothing in a formula knows whether a container has a lid. Only the sentence does. Before computing any surface area, write down the list of faces you are counting — and for a cylinder, say whether you are counting two circles, one, or none.

13. An open-topped glass aquarium is a rectangular prism 80 cm long, 35 cm wide and 45 cm deep. It has no lid.



(a) List the faces that must be made of glass. Then find the total area of glass required, in cm^2.

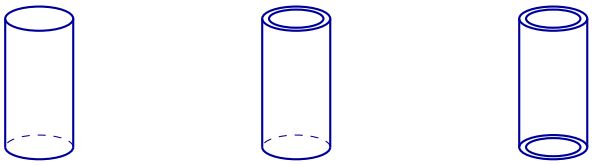
(b) Find the capacity of the aquarium in litres when it is filled to the brim.

(c) A glass lid is added. State the new total area of glass, and the percentage increase, to the nearest tenth of a percent.

(d) Explain why parts (a) and (b) need two different units, and why neither answer could be reported in the other's unit.

14. Three cylindrical objects all have radius 4.0 cm and height 30.0 cm. Object I is a sealed can. Object II is a cup, open at the top only. Object III is a length of drainpipe, open at *both* ends; only its outer surface is to be found.

all three: $r = 4.0$ cm, $h = 30.0$ cm



I — sealed can II — open top only III — open both ends

(a) For each object, write the surface-area expression first — stating how many circles it includes — then evaluate it to the nearest tenth.

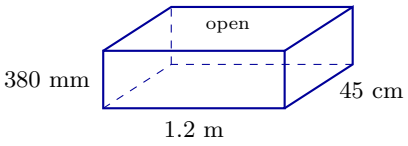
(b) The three answers differ only by whole circles. State the area of one circle, to the nearest tenth, and check that your three answers differ by exactly that much, then twice that much.

(c) All three objects have the same volume. State it, to the nearest tenth, and explain why the wording about lids changed the surface area but could not change the volume.

Part F — Reconciling Units Before You Start

A formula cannot be trusted to notice that one dimension is in metres and another in millimetres. Convert everything to a single unit first*, and record which unit you chose.*

15. A planter box, open at the top, is 1.2 m long, 45 cm wide and 380 mm deep.



(a) Convert all three dimensions to centimetres, then find the volume of soil the box holds, in cm^3, and convert it to litres.

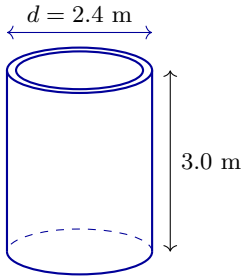
(b) Redo the volume working entirely in metres, giving the answer in m^3. Check that it agrees with (a).

(c) Find the area of timber needed to build the box (five faces — no top), first in cm^2 and then in m^2.

(d) The conversion from cm^2 to m^2 is not the same number as the conversion from cm^3 to m^3. State both, and explain the difference in one sentence.

Part G — What It Is Made Of Versus What It Holds

16. An open-topped cylindrical water tank is to be built from sheet steel. It has a base diameter of 2.4 m and a height of 3.0 m. Sheet steel costs \$47.50 per square metre, and the tank has a steel base but no lid.



(a) Find the area of steel needed, to the nearest hundredth of a square metre. Say exactly which surfaces you counted.

(b) Find the cost of the steel, to the nearest cent.

(c) Find the capacity of the tank in m^3 and in litres, each to the nearest tenth.

(d) Find the cost of the steel per litre of capacity, to the nearest tenth of a cent, and say in one sentence why a builder would want this figure rather than the total cost.

Part H — Error Analysis

Each student below made exactly one error. Name it in words, give the corrected work in full, and state the check that would have caught it before the answer was handed in.

17. **The student who used the diameter as the radius.** A cylindrical tin has diameter 14.0 cm and height 20.0 cm. Asked for its volume to the nearest tenth, a student wrote

$$V = \pi r^2 h = \pi(14.0)^2(20.0) = 12\,315.0 \text{ cm}^3$$

(a) Name the error and give the correct volume, to the nearest tenth.

(b) Divide the student's answer by the correct one. The result is a whole number. Explain why mistaking d for r always multiplies a cylinder's volume by exactly that number.

(c) State a one-line check, available before any arithmetic is done, that would have caught this.

18. **The student whose unit gave the game away.** Asked for the *surface area* of a cube with edge 9.0 cm, a student wrote

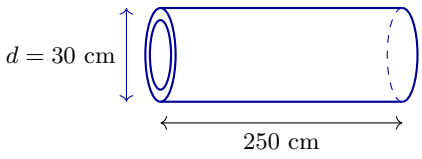
$$9.0 \times 9.0 \times 9.0 = 729 \text{ cm}^3$$

(a) Without doing any of the arithmetic again, point to the single symbol in the student's line that proves the answer cannot be a surface area, and say what it proves.

(b) Give the correct surface area, with its unit. Then state the rule connecting the number of dimensions multiplied together to the exponent on the unit.

19. **The student who capped an open pipe.** A length of drainpipe is open at both ends. Its outside diameter is 30 cm and it is 250 cm long. Asked for the area of the outside surface to be coated, a student wrote

$$SA = 2\pi r^2 + 2\pi r h = 2\pi(15)^2 + 2\pi(15)(250) = 24\,975.7 \text{ cm}^2$$



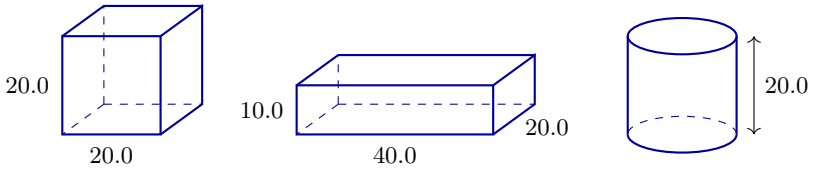
(a) Name the error, and say which term in the student's expression should not be there.

(b) Give the correct area to be coated, to the nearest tenth, and state by how much the student over-counted.

(c) Rewrite the question's wording so that the student's expression *would* have been correct, changing as little as possible.

Part I — Stretch: Same Volume, Different Amount of Material

20. Three containers each hold exactly 8000 cm^3. Container A is a cube of edge 20.0 cm. Container B is a rectangular prism 40.0 cm by 20.0 cm by 10.0 cm. Container C is a closed cylinder of height 20.0 cm. All three are closed on every face.



A — cube B — rectangular prism C — cylinder, r =?

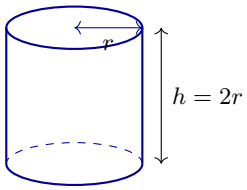
(a) Verify that A and B each hold 8000 cm^3, then find the radius of C, to the nearest hundredth of a centimetre.

(b) Find the surface area of each container, to the nearest tenth.

(c) Rank the three by the amount of material used. State which container is cheapest to build and by how many cm^2 it beats the most expensive.

(d) Container B is the same volume as container A but uses more material. Describe, in terms of how *stretched out* a shape is, the rule these three containers illustrate — and predict, without calculating, whether a cube of edge 20.0 cm or a 80.0 cm by 10.0 cm by 10.0 cm box would need more material.

21. A cylinder is called **equilateral** when its height equals its diameter, so $h = 2r$.



(a) Write SA and V for an equilateral cylinder in terms of r alone, simplifying each as far as possible.

(b) Form the ratio $SA : V$ and simplify it. The answer should contain no π. State what it says about large containers compared with small ones of the same shape.

(c) An equilateral cylinder holds 1000 cm^3. Find r, then h, then SA, each to the nearest tenth.

(d) A cube also holds 1000 cm^3. Find its surface area, compare it with your answer to (c), and say which shape a manufacturer should choose.

Lesson 4 — Extra Practice

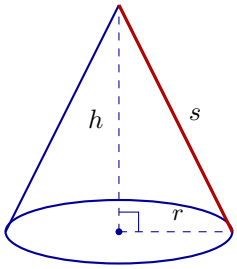
Surface Area and Volume of Pyramids and Cones

Part A — Two Heights, One Right Triangle

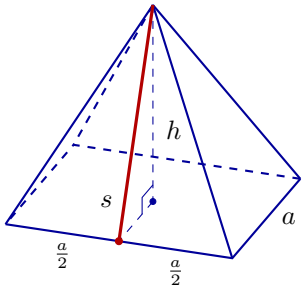
Every question in this set turns on one distinction. The ***perpendicular height*** *h runs from the apex straight down to the* centre *of the base. The* ***slant height*** *s runs from the apex down the* surface *of the solid to the* nearest *point of the base edge.* ***Surface area uses s; volume uses h.*** *They are never equal, and Pythagoras links them. Throughout: use the π key — **never** 3.14 — carry full precision, and round once at the very end, to the precision each question names. Every answer carries its unit.*

1. Each solid below has its perpendicular height drawn as a dashed line and its slant height drawn as a heavy line. For each solid state (i) which drawn segment is h and which is s, and where each one ends; (ii) the two legs and the hypotenuse of the right triangle the two heights belong to; (iii) the Pythagorean equation linking them.

(a) right circular cone



(b) right square-based pyramid



(c) In the pyramid the horizontal leg is $\frac{a}{2}$, not a. Explain from the figure why, and say what the number $\sqrt{h^2 + a^2}$ would be measuring instead.

2. Find the missing height in each solid. Give an exact value where one exists, otherwise round to the nearest tenth. State each time whether you are finding a perpendicular height or a slant height.

(a) a cone with $r = 9$ cm and $s = 41$ cm; find h

(b) a cone with $r = 11.0$ m and $h = 5.0$ m; find s

(c) a square pyramid with base edge 16 in and $h = 15$ in; find s

(d) a square pyramid with base edge 10 cm and $s = 13$ cm; find h

3. Two measurement reports are impossible. Explain each one.

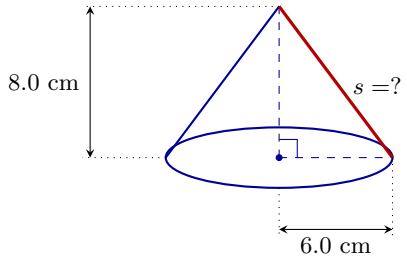
(a) "A cone of perpendicular height 10 cm and slant height 8 cm." Show that no such cone exists, and state the inequality that s must always satisfy in *every* cone.

(b) "A square pyramid of base edge 24 cm and slant height 12 cm." Show that this one is impossible too, and say exactly what shape these two numbers do describe.

Part B — Surface Area Takes s, Volume Takes h

In each question here you are given only one *of the two heights. Find the other before you start, and name which height each formula is using.*

4. A solid cone has base radius 6.0 cm and perpendicular height 8.0 cm.

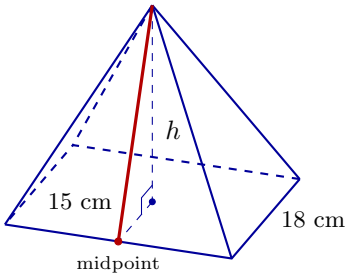


(a) Determine the total surface area, to the nearest tenth of a square centimetre.

(b) Determine the volume, to the nearest tenth of a cubic centimetre.

(c) Your two answers are the same number. Explain why that does *not* mean the surface area and the volume are equal, and show by changing the height to 9.0 cm that the coincidence is an accident of these particular dimensions.

5. A right pyramid has a square base of edge 18 cm and a slant height of 15 cm.



(a) Determine the total surface area, to the nearest square centimetre.

(b) Determine the volume, to the nearest cubic centimetre.

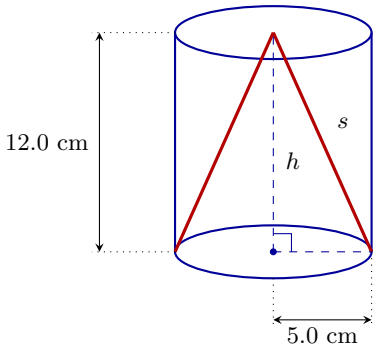
(c) State which height each of (a) and (b) used, and calculate what answer (b) would have given if the slant height had been used by mistake. By what percent would that answer be too large?

6. A conical coffee filter has a base diameter of 14.0 cm and a slant height of 25.0 cm. How many millilitres of water does it hold when filled to the brim? ($1\ \text{cm}^3 = 1\ \text{mL}$.) Answer to the nearest millilitre, and name the height your formula uses.

Part C — Exactly One Third

A cone holds exactly one third of the cylinder on the same base with the same height, and a pyramid exactly one third of the prism. Derive the ratio in these questions — do not quote it.

7. A cone of base radius 5.0 cm and perpendicular height 12.0 cm sits inside the smallest cylinder that contains it.



(a) Calculate each volume from its own formula, exactly in terms of π and to the nearest tenth of a cubic centimetre, then form the ratio $V_{\text{cone}} : V_{\text{cylinder}}$.

(b) Show with algebra that this ratio is $\frac{1}{3}$ for *every* radius and *every* height.

(c) Determine the volume of the empty space left between the cone and the cylinder, to the nearest tenth, and state it as a fraction of the cylinder.

(d) Is the cone's *curved surface* also one third of the cylinder's curved surface? Calculate both and decide. Say which height each one needs.

8. A solid block is a square prism with base edge 9.0 cm and height 20.0 cm. The largest possible pyramid on the same square base and with the same height is carved from it; the pyramid is kept and the rest is waste.

(a) Find the volume of the block.

(b) Find the volume of the pyramid *without* using the pyramid formula.

(c) Find the volume of the waste.

(d) Explain why the waste is exactly twice the pyramid.

Part D — Running the Formulas Backwards

Here the volume or the area is the given, and a length is the unknown. Rearrange first, substitute second, and keep the unrounded value for any step that follows.

9. A conical paper cup holds exactly 250 mL when filled to the brim, and the radius of its rim is 4.0 cm.

(a) Determine the perpendicular height of the cup, to the nearest tenth of a centimetre.

(b) Determine the slant height, to the nearest tenth of a centimetre.

(c) The cup has no base. Determine the area of paper needed, to the nearest square centimetre.

10. A monument is a right pyramid on a square base. Its volume is 1200 m^3 and its perpendicular height is 25 m.

(a) Determine the length of a base edge.

(b) Determine the slant height, to the nearest tenth of a metre.

(c) Determine the total area of the four triangular faces, to the nearest square metre.

11. The curved surface of a cone has an area of exactly 96π cm^2, and the base radius is 6.0 cm.

(a) Determine the slant height.

(b) Determine the perpendicular height, to the nearest tenth of a centimetre.

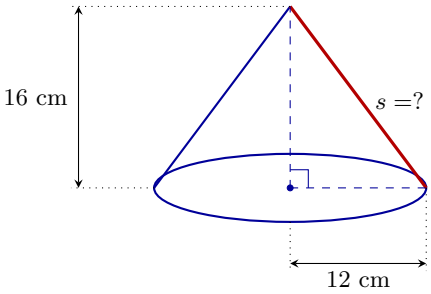
(c) Determine the volume, to the nearest tenth of a cubic centimetre.

(d) Determine the total surface area, exactly in terms of π.

Part E — Lateral Area or Total Surface Area? The Wording Decides

The curved surface of a cone alone is πrs; the whole outside of a solid cone is $\pi rs + \pi r^2$. Nothing in the numbers tells you which is wanted — only the sentence does. In each part, ***name the formula you used and quote the words that made you choose it.***

12. Three objects are all cones of base radius 12 cm and perpendicular height 16 cm.



(a) A party hat rolled from one piece of card and open at the bottom. How much card?
(b) A solid clay cone dipped entirely in glaze. How much glazed surface?
(c) A tent with canvas walls and a sewn-in circular nylon groundsheet. How much canvas, and how much nylon?
Answer each to the nearest tenth of a square centimetre.

13. A cylindrical grain bin is topped by a conical steel roof of base diameter 8.0 m and slant height 5.0 m.

(a) How many square metres of steel sheeting does the roof need, to the nearest tenth?

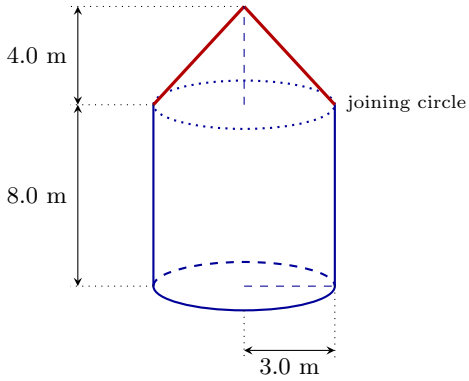
(b) Explain why the circle at the bottom of the roof is not part of that sheeting.

(c) How much space does the roof enclose above the bin, to the nearest tenth of a cubic metre?

Part F — Composite Solids: The Joining Face Belongs to Neither Surface

When two solids are glued together, the face they share is inside *the new solid. It is in neither exterior surface, and counting it is the single commonest mistake in this lesson.*

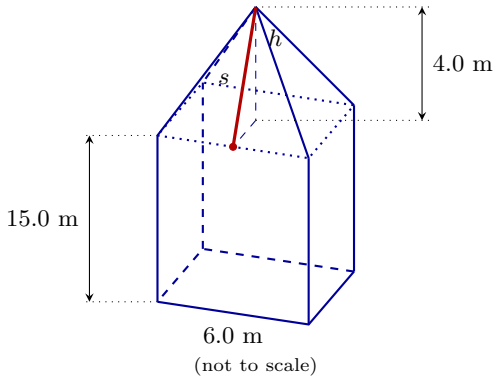
14. A grain silo is a cylinder of radius 3.0 m and height 8.0 m, topped by a cone of the same radius and perpendicular height 4.0 m. The silo stands on the ground.



(a) Determine the total volume, to the nearest tenth of a cubic metre.
(b) The curved wall of the cylinder and the curved surface of the cone are to be painted. Determine the area to be painted, to the nearest tenth of a square metre.

(c) Two circles of this solid appear in no painted surface. Name them both and give a different reason for each. State the area of the one that a careless student would add in.

15. A church tower is a square prism of base edge 6.0 m and height 15.0 m, capped by a right pyramid on the same square base with a perpendicular height of 4.0 m.



(a) Determine the total volume of the tower.
(b) The four tower walls and the four roof faces are to be clad. Determine the area to be clad.
(c) Two squares of area 36 m^2 lie in this solid but on no clad surface. Name them both.

Part G — Error Analysis: Four Students, Four Different Failures

For each student: name the error in one sentence, say which height or which face they should have used, and give the correct answer.

16. Find the mistake.

(a) Dana — a cone with $r = 3$ cm, $h = 4$ cm and $s = 5$ cm.
Dana writes $V = \frac{1}{3}\pi r^2 s = \frac{1}{3}\pi(3)^2(5) = 15\pi \approx 47.1$ cm^3.

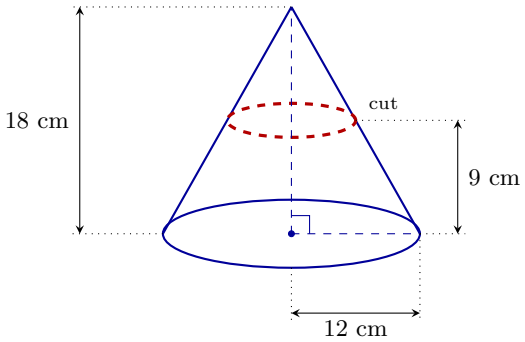
(b) Ellis — a square pyramid with base edge 16 cm and perpendicular height 6 cm.
Ellis writes $s = \sqrt{6^2 + 16^2} = \sqrt{292} \approx 17.1$ cm.

(c) Farid — a square pyramid with base edge 10 cm and perpendicular height 12 cm.
Farid writes $V = (10)^2(12) = 1200$ cm^3.

(d) Gwen — a cylinder of radius 5 cm and height 10 cm with a cone of the same radius and perpendicular height 12 cm on top.
Gwen writes
$SA = 25\pi + 100\pi + 25\pi + 65\pi = 215\pi \approx 675.4$ cm^2, the four terms being the bottom circle, the cylinder wall, the top of the cylinder, and the cone.

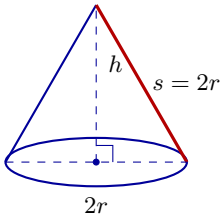
Part H — Stretch

17. A conical funnel has base radius 12 cm and perpendicular height 18 cm. A plane parallel to the base cuts it 9 cm above the base, and the small cone above the cut is removed.



(a) Explain why the small cone's base radius is 6 cm.
(b) Find the volume of the whole cone and of the small cone, each exactly in terms of π, using the one-third rule twice.
(c) Find the volume of the piece that remains, exactly and to the nearest tenth of a cubic centimetre.
(d) What fraction of the original volume was removed? Explain why it is not $\frac{1}{2}$.

18. A cone has a slant height exactly equal to its base *diameter*: $s = 2r$.



(a) Show that $h = r\sqrt{3}$.
(b) Show that the total surface area is exactly $3\pi r^2$, and that the curved surface is exactly twice the base.
(c) One such cone has a total surface area of 300 cm^2. Find r, to the nearest tenth of a centimetre.
(d) Find that cone's volume, to the nearest tenth of a cubic centimetre.

19. A cone and a right square-based pyramid have the same perpendicular height, 15.0 cm, and equal volumes. The cone's base radius is 8.0 cm.

(a) Determine the pyramid's base edge, to the nearest tenth of a centimetre.

(b) A classmate guesses the base edge must be 16.0 cm, "because that is the diameter." Without redoing (a), explain why 16.0 cm is too big, and state by how many square centimetres the two base areas would differ.

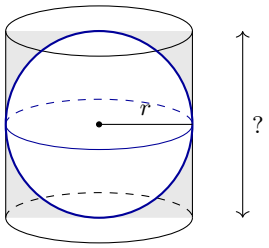
Lesson 5 — Extra Practice

Surface Area and Volume of Spheres and Hemispheres

Part A — Two Thirds of the Cylinder That Just Contains It

*Throughout this set use **exact** π — never 3.14 — carry full precision through every intermediate step, and **round once, at the end**, to the precision each question names. **Every answer carries its unit.** When a diameter is given, halving it is the first line of the solution, not an afterthought. Formulas:* $V_{sphere} = \frac{4}{3}\pi r^3$ *and* $SA_{sphere} = 4\pi r^2$.

1. A sphere of radius r sits inside the smallest cylinder that will contain it: the sphere touches the flat top, the flat bottom and the curved wall all at once. The shaded region is the space left over.



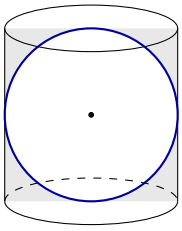
(a) State the cylinder's radius and its height in terms of r, and explain in one sentence why the height must be what you say it is.

(b) Write the cylinder's volume in terms of r alone, simplified as far as it will go.

(c) Mika's overflow experiment showed that the sphere fills exactly **two thirds** of this cylinder. Starting from that fact and your answer to (b), **derive** $V_{\text{sphere}} = \frac{4}{3}\pi r^3$. Show every line — do not quote the formula.

(d) **The leftover third.** Write the volume of the shaded space in terms of r, once by subtraction and once from the two-thirds statement, and confirm the two agree. What fraction of the cylinder is it?

2. A cylindrical gift tin holds a spherical glass ornament snugly — the ornament touches the lid, the base and the curved wall. The tin's interior volume is exactly 1620 cm^3.



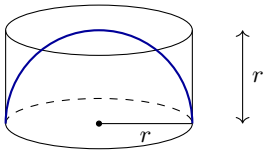
tin: 1620 cm^3

(a) **Using the ratio only**, and without finding the radius, state the volume of the ornament and the volume of air around it. Justify each in one line.

(b) Now do it the long way: find the radius from the tin's volume (to the nearest tenth of a centimetre), then substitute into $V = \frac{4}{3}\pi r^3$, and confirm your answer to (a).

(c) One route needed no calculator at all and the other needed a cube root. Say which piece of information made the short route possible, and state the one condition a cylinder must satisfy before the two-thirds shortcut may be used.

3. The same two-thirds idea works one level down. A **hemisphere** of radius r is set inside a cylinder of radius r whose height is r, so that it just fits.



(a) Show, algebraically, that the hemisphere fills exactly two thirds of *this* cylinder too. Start from $V_{\text{hemisphere}} = \frac{1}{2} \cdot \frac{4}{3}\pi r^3$.

(b) A hemispherical bowl of radius 9.0 cm is filled level with its rim and poured into a cylindrical glass of radius 9.0 cm and height 9.0 cm. How far up the glass does the water rise? Answer to the nearest tenth of a centimetre — and get there **without computing a single volume**.

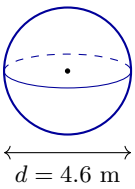
(c) The radius 9.0 cm never mattered. Explain why the rise in (b) is always the same fraction of the glass's height, whatever the common radius is, and name that fraction.

Part B — The Diameter Is Not the Radius

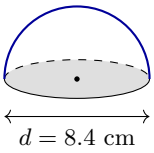
*Every question in this part hands you a **diameter**. Halve it first, write the radius down, and only then substitute. Using d where r belongs multiplies a surface area by 4 and a volume by 8.*

4. Calculate each of the following. Give the exact value in terms of π *and* the decimal value to the nearest tenth.

(a) The surface area and the volume of a spherical weather balloon of diameter 4.6 m.



(b) The volume of a solid hemispherical paperweight of diameter 8.4 cm, and the area of glass to be polished if **every** outside surface of the paperweight is polished, flat face included.



5. Sphere P has a **diameter** of 14.0 cm. Sphere Q has a **radius** of 14.0 cm.

(a) **Before calculating anything**, predict how many times larger Q's volume is than P's, and justify the prediction from the formula.

(b) Now find both volumes exactly in terms of π and to the nearest tenth of a cubic centimetre. Check your prediction.

(c) State in one sentence what this pair of spheres tells you about the cost of reading a diameter as a radius.

6. Which has the greater surface area: a **sphere** of diameter 20.0 cm, or a **solid hemisphere** of diameter 24.0 cm whose curved surface *and* flat base are both counted?

(a) Answer by working both surface areas **exactly, in terms of** π, and comparing the coefficients. Do not reach for a calculator.

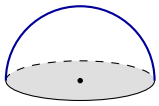
(b) By what percentage is the larger surface area greater than the smaller? Because you kept π exact, this percentage comes out as a whole number — explain why π cancels.

Part C — A Hemisphere's Flat Base: The Wording Decides

*A hemisphere's curved surface is $2\pi r^2$ and its flat circular base is πr^2, so the total is $3\pi r^2$. Which of the two a question wants is settled by **the wording**, not by habit. State the surfaces you are counting, every time.*

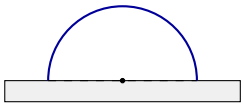
7. Three different objects are each a hemisphere of diameter 21.0 cm. For each part, (i) state which surfaces you are counting and why the wording forces that choice, (ii) write the formula you will use, and (iii) calculate the area to the nearest tenth of a square centimetre.

(a) A *solid wooden* hemisphere, to be varnished all over



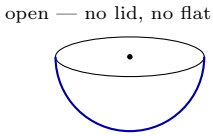
flat face is wood, and is varnished
$d = 21.0$ cm

(b) A hemispherical *dome* sitting on a flat roof, the outside to be painted



roof deck
the dome has no floor
$d = 21.0$ cm

(c) A hemispherical *bowl*, open at the top, the outside to be glazed



$d = 21.0$ cm

(d) Two of your three answers are identical. Say which two, explain why the objects are so different yet the arithmetic is the same, and then write the general rule in one sentence beginning "Count the flat circular base only when…".

8. A hemispherical observatory dome is built with **no floor**. Its curved surface area is 150 m^2.

(a) Determine the dome's radius, to the nearest tenth of a metre. Start by writing down which surface-area expression the wording tells you to use.

(b) Determine the floor area the dome encloses — the circle it stands on — to the nearest tenth of a square metre. Work from the *unrounded* radius.

(c) Your answer to (b) is exactly half of 150. Prove that a hemisphere's curved surface is *always* exactly twice its base, for every radius, and say why this gives a ten-second check on part (b).

Part D — Running the Formulas Backwards

*Given a volume, a cube root recovers the radius; given a surface area, a square root does. Rearrange the formula in symbols **first**, substitute second, and keep the unrounded radius on your calculator for anything that follows.*

9. A spherical helium balloon holds 12 500 cm^3 of gas.

(a) Rearrange $V = \frac{4}{3}\pi r^3$ to isolate r. Write the result both as a radical and as a power.

(b) Determine the balloon's radius to the nearest tenth of a centimetre.

(c) Determine its diameter to the nearest tenth of a centimetre, taking care that the doubling happens *before* the rounding.

10. A spherical steel buoy has a surface area of 92.0 m^2. Rearrange $SA = 4\pi r^2$ to isolate r, then determine the buoy's radius to the nearest tenth of a metre.

11. A spherical propane tank has a surface area of 95.0 m^2.

(a) Determine its volume to the nearest tenth of a cubic metre, via the radius. Carry the *unrounded* radius into the volume step.

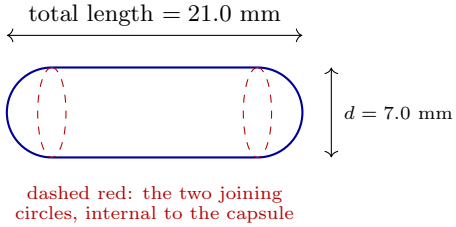
(b) Repeat the volume calculation using the radius *rounded to the nearest tenth* before substituting. State the size of the disagreement and which answer is the right one to report.

(c) Show algebraically that $V = \frac{1}{3}\,SA \cdot r$ for every sphere, then use it to obtain the volume in part (a) in a single multiplication.

Part E — Composite Solids: Joining Circles Belong to Neither Surface

*When two solids are glued together, the circle where they meet is **inside** the finished object. It is counted in **neither** surface area — and it is the single most common slip in this part. Volumes, by contrast, simply add.*

12. A pharmaceutical capsule is a cylinder with a hemisphere on each end. Its total length is 21.0 mm and its diameter is 7.0 mm. The dashed circles mark where the caps meet the cylinder.



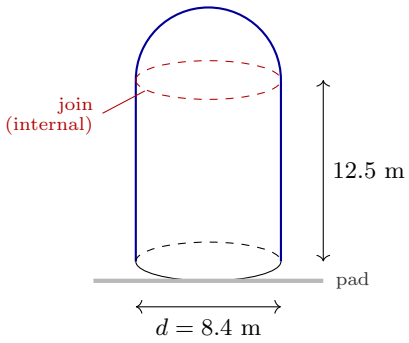
(a) Find the length of the cylindrical middle. Then list every surface that is part of the capsule's outside and every circle that is **not**.

(b) Determine the capsule's volume, exactly in terms of π and to the nearest tenth of a cubic millimetre. (The two hemispherical ends can be recombined — say how.)

(c) Determine the capsule's surface area, exactly in terms of π and to the nearest tenth of a square millimetre.

(d) A student adds $2\pi r^2$ to the surface area, "one circle for each end of the cylinder". By how many square millimetres is their answer too large, and in one sentence why is that area not part of the capsule's surface?

13. A grain silo is a cylinder of diameter 8.4 m and height 12.5 m, capped by a hemispherical roof of the same diameter. The silo stands on a concrete pad.

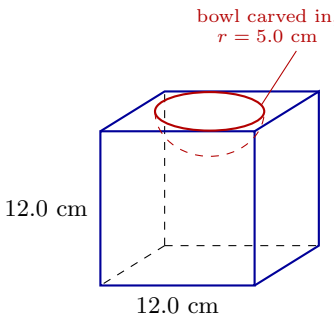


(a) Determine the silo's total interior volume, exactly in terms of π and to the nearest tenth of a cubic metre.

(b) The whole exterior is to be painted *except* the circular base resting on the pad. Determine the area to be painted, exactly in terms of π and to the nearest tenth of a square metre.

(c) Exactly two circles of area πr^2 are absent from your answer to (b). Name them both, and say which one is excluded by *geometry* and which by the *wording* of the question.

14. A hemispherical bowl of radius 5.0 cm is carved down into the centre of the top face of a solid wooden cube of edge 12.0 cm.



(a) Determine the volume of wood remaining, exactly in terms of π and to the nearest tenth of a cubic centimetre.

(b) Determine the total surface area of the carved block, exactly in terms of π and to the nearest tenth of a square centimetre. Remember that the circle of wood removed from the top face is no longer a surface.

(c) Carving the bowl *increased* the surface area by exactly πr^2. Show this, then show that **gluing** a hemisphere of the same radius onto the top face instead would have increased it by exactly πr^2 as well. Explain the coincidence.

Part F — Scaling: Where the 4 and the 8 Come From

*The scaling rule — a linear factor is **squared** for areas and **cubed** for volumes — is not a separate fact about conversions. It is the exponents in the formulas, read out loud. Reason it out first; check it numerically second.*

15. Sphere S has radius 5.0 cm. Sphere T has radius 10.0 cm.

(a) **Without computing any numbers**, use the exponents in $SA = 4\pi r^2$ and $V = \frac{4}{3}\pi r^3$ to state the factor by which the surface area grows and the factor by which the volume grows. Show the algebra by replacing r with $2r$.

(b) Now compute all four quantities, exactly in terms of π and to the nearest tenth, and verify both factors.

(c) A sphere's radius is multiplied by k. State the factor for the surface area and the factor for the volume, and say in one sentence how this is the *same* rule as "1 ft = 0.3048 m becomes 1 $\text{ft}^2 = 0.092\,903\ \text{m}^2$ and 1 $\text{ft}^3 = 0.028\,316\,8\ \text{m}^3$".

(d) A sphere's volume is multiplied by 27. By what factor was its radius multiplied, and by what factor its surface area? Show the reasoning, not a guess.

16. A chocolate truffle has a diameter of 5.0 cm. A giant display truffle has a diameter of 15.0 cm.

(a) Using the scaling rule alone — no volume formula — state how many ordinary truffles' worth of chocolate the display truffle contains.

(b) Verify by computing both volumes exactly in terms of π and to the nearest tenth of a cubic centimetre.

(c) The truffles are coated in foil. How many times as much foil does the display truffle need? State the factor, then verify with both surface areas.

Part G — Capacity, Packing, and Displacement

A volume in m^3 becomes a capacity in litres through $1\ m^3 = 1000\ L$; a volume in cm^3 becomes millilitres through $1\ cm^3 = 1$ mL. In the last two questions the π cancels before you ever reach a calculator — notice when it does.

17. A spherical water tank has a diameter of 3.6 m.

(a) Determine the tank's volume, exactly in terms of π and to the nearest hundredth of a cubic metre.

(b) Report the tank's capacity in litres, to the nearest litre.

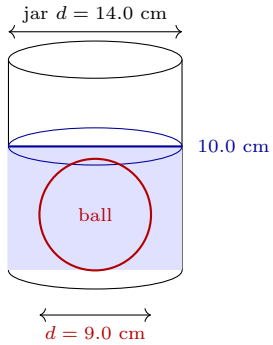
(c) A household draws 310 L per day. For how many **whole** days does one full tank last? Justify why the answer is rounded *down* rather than to the nearest whole number.

18. Spherical scoops of ice cream 6.0 cm in diameter are packed into a cylindrical tub of diameter 15.0 cm and height 12.0 cm. Assume the ice cream is pressed down so no air is trapped, so this is purely a comparison of volumes.

(a) How many **whole** scoops does the tub hold? Work exactly in terms of π and state the volume left over, to the nearest tenth of a cubic centimetre.

(b) The whole-number part of this answer needed no decimal approximation at all. Explain what cancels, and say what that tells you about comparing two volumes of the same kind.

19. A cylindrical jar of diameter 14.0 cm contains water to a depth of 10.0 cm. A solid steel ball of diameter 9.0 cm is lowered in and comes to rest on the bottom, completely below the surface. No water is spilled.



(a) Determine the rise in the water level, to the nearest tenth of a centimetre. Set the ball's volume equal to the volume of a *cylindrical slab* of water of the jar's radius, and solve for its height.

(b) The π disappears from the equation before any arithmetic happens. Show where it goes, and state the exact fractional value of the rise.

(c) Confirm that the ball really was completely submerged after the rise, as the question assumed. State the new depth and compare it with the ball's diameter.

Part H — Error Analysis: Three Students, Three Different Failures

Each student below made exactly one error. Name it in words, give the corrected work, and — where one exists — give the reasonableness check that would have caught it before the answer was handed in.

20. A solid sphere has a diameter of 16.0 cm.

(a) **Devon** wrote $V = \frac{4}{3}\pi(16.0)^3 = 17\,157.3$ cm^3. Name the error, give the correct volume to the nearest tenth of a cubic centimetre, and state the exact factor relating Devon's answer to yours — then say where that factor comes from.

(b) The sphere fits exactly inside a cube of edge 16.0 cm. Compute that cube's volume and use it to write, in one sentence, the reasonableness check that kills Devon's answer on sight.

(c) **Fern** halved the diameter correctly but then wrote $V = \frac{4}{3}\pi(8.0)^2 = 268.1$ cm^3. Name the error, and say what unit her expression actually produces — and why the unit alone is enough to convict her.

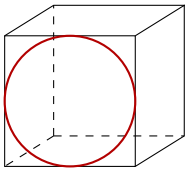
(d) **Idris** was asked for the area of a hemispherical dome of diameter 16.0 cm that is built **with no floor**, and answered $3\pi r^2 = 603.2$ cm^2. Name the error, give the correct area to the nearest tenth, and state exactly how much area was wrongly included.

Part I — Stretch: The Largest Sphere That Fits

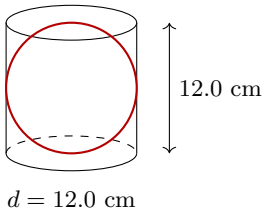
A sphere packed into a box always leaves gaps, and how big the gaps are depends entirely on the shape of the box. Work in terms of π for as long as you can; the fractions turn out to be exact.

21. The largest possible sphere is placed inside each of the three containers below. For each one: state the sphere's radius, give the container's volume and the sphere's volume exactly in terms of π, and give the fraction of the container the sphere occupies — as an exact fraction and as a percentage to the nearest tenth.

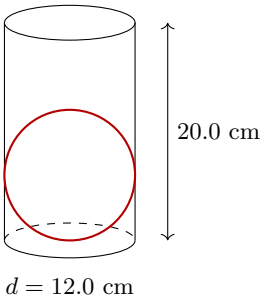
(a) cube, edge 12.0 cm



(b) cylinder, $h = 12.0$ cm



(c) cylinder, $h = 20.0$ cm



(a) the cube

(b) the cylinder with $h = 12.0$ cm

(c) the cylinder with $h = 20.0$ cm

(d) Show that for a cube of *any* edge length the fraction in (a) is exactly $\dfrac{\pi}{6}$, so the edge length 12.0 cm was never needed.

(e) The same sphere occupies a different fraction of each container. Rank the three from best fit to worst, explain why making the cylinder *taller* makes the fit *worse*, and state which cylinder is the best possible one — connecting your answer to Part A.

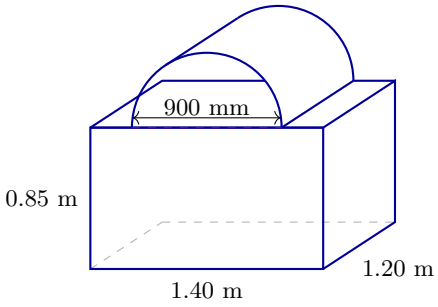
Lesson 6 — Extra Practice

Measurement Unit Review

This is the last set of the course, and every question in it deliberately crosses lesson boundaries: a reading becomes a conversion, a conversion becomes a volume, a volume becomes a cost. Throughout: ***exact*** π***, never*** 3.14*;* ***carry full precision through every intermediate step and round once, at the end****, to the precision each question names; every answer carries its unit, with the right exponent. Where a height is used,* ***say which height it is*** *— the perpendicular height* h *or the slant height* s*. Where a hemisphere appears,* ***say whether its flat circular base is counted****, and why the wording forces that choice. Write every conversion as a unit factor, so the unwanted unit visibly cancels.*

Part A — One Solid, Every Question

1. **The bread oven.** A backyard wood-fired bread oven is built in two pieces. The lower piece is a solid concrete plinth: a right rectangular prism 1.40 m wide, 1.20 m deep and 0.85 m high. The upper piece is the baking chamber: a right prism whose base is a **semicircle** of diameter 900 mm, laid on its flat rectangular face on top of the plinth and running the full 1.20 m depth, so its two semicircular ends sit directly above the front and back faces of the plinth. The red dashed segment marks the front edge of the flat rectangular face where the chamber meets the plinth — that whole face is *internal*.



(a) Convert the chamber's diameter to metres with a unit factor, and state the radius r you will substitute.

(b) Determine the **total volume** of concrete and chamber together — the plinth plus the half-cylinder — to the nearest hundredth of a cubic metre. Name the base area you used for the semicircular prism.

(c) Determine the **capacity of the baking chamber alone**, in litres, to the nearest litre. ($1\ \text{m}^3 = 1000$ L.)

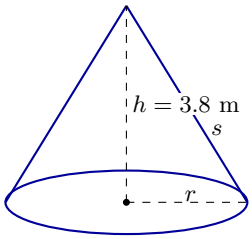
(d) Determine the **total outside surface area** of the finished oven, to the nearest hundredth of a square metre. The oven stands on the ground, so the bottom of the plinth is not counted; the rectangular face where the chamber meets the plinth is internal and is counted on *neither* piece. List every face you keep and every face you drop.

(e) Weatherproof stucco costs \$38.50 per square metre and is applied to every surface you counted in (d). Find the cost to the nearest cent, using the **unrounded** area.

(f) A second oven of the same design is to have a baking chamber holding exactly 500 L, with the same 900 mm diameter. How deep must that oven be — that is, what chamber length is needed? Give it to the nearest centimetre.

(g) A stainless flue is fitted: a cylinder of diameter 20.0 cm and height 40.0 cm, capped by a cone of base diameter 28.0 cm and *perpendicular* height 10.5 cm. Find the slant height of the cone, then the area of stainless sheet needed for the curved wall of the cylinder and the curved surface of the cone (no ends, no cap underside). Give the answer in square centimetres to the nearest hundredth, and then in square metres.

2. **The salt pile.** Road salt is stockpiled in a cone. A tape run around the base of the pile gives its circumference as 26.4 m, and a measuring pole gives its perpendicular height as 3.8 m.



$C = 26.4$ m

(a) The radius was never measured. Rearrange $C = 2\pi r$ and find r, and say why you must *not* round it before going on.

(b) Determine the volume of salt in the pile, to the nearest tenth of a cubic metre. Name the height your formula uses.

(c) Express that volume in cubic feet, to the nearest cubic foot, building the cubic factor from 1 ft = 0.3048 m rather than quoting one.

(d) The pile is to be covered with tarpaulin. Find the slant height s, then the **lateral** area of the cone, to the nearest tenth of a square metre. Tarps come in 6.0 m × 8.0 m sheets; how many are needed?

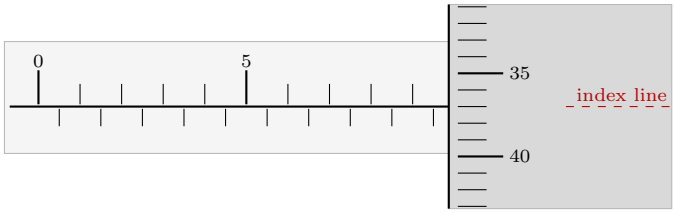
(e) Salt has a density of 1.20 tonnes per cubic metre. Find the mass of the pile to the nearest tenth of a tonne.

(f) Trucks remove salt until only 50.0 m^3 remains, and the pile keeps its 3.8 m height. What would a tape run around the base of the smaller pile now read, to the nearest tenth of a metre?

Part B — Measure, Convert, Compute

A reading is the start *of a calculation, never the end of one. Read the instrument, name its precision, carry full precision into the formula, and round once.*

3. A steel ball bearing is measured with a micrometer. The sleeve carries a mark every 0.5 mm — whole millimetres above the index line, half millimetres below it — and the thimble reads to 0.01 mm.



Sleeve: a mark every 0.5 **mm**
Thimble: 0.01 **mm per division**

(a) State the course reading and the fine reading separately (show the multiplication for the fine reading), then the diameter. Write it to the micrometer's own precision.

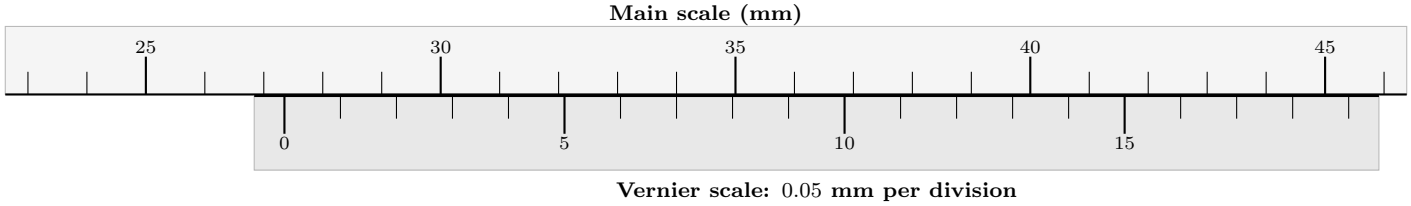
(b) Determine the ball's volume, to the nearest tenth of a cubic millimetre, and then in cubic centimetres to the nearest thousandth. Show the cubic unit factor.

(c) Determine the ball's surface area in square millimetres and in square centimetres, each to the nearest hundredth.

(d) Steel has a density of 7.85 g/cm^3. Find the mass of one bearing to the nearest hundredth of a gram, and then the greatest whole number of bearings in a 1.00 kg bag.

(e) A student reads the sleeve as "9" instead of "9.5" and reports 9.37 mm. Without recomputing from scratch, state the factor by which that student's *volume* would be wrong, to three decimal places, and explain why the factor is not simply 9.37/9.87.

4. A solid brass drawer knob is a **hemisphere**. Its diameter is measured with a vernier caliper reading to 0.05 mm.



(a) State the course reading, the fine reading (with the multiplication), and the total, written to the caliper's own precision. Then state the radius.

(b) Determine the volume of brass in one knob, in cubic millimetres to the nearest hundredth and in cubic centimetres to the nearest thousandth.

(c) The **curved** surface is chrome-plated; the flat circular face is glued to the drawer and is *not* plated. State which formula that wording selects, and find the plated area in square millimetres to the nearest hundredth.

(d) Plating costs \$185.00 per square metre. Find the cost of plating a batch of 250 knobs, to the nearest cent. Show the $\text{mm}^2 \to \text{m}^2$ factor and say where its exponent comes from.

(e) Brass has a density of 8.47 g/cm^3. How many whole knobs can be cast from 1.00 kg of brass, if none is wasted?

5. **A referent against a tape.** Kavi estimates the top of a cylindrical rain barrel by laying his hand-spans across it: it takes exactly 3 spans, and he takes his span to be 22 cm. A tape is then run *around* the barrel and reads a circumference of 1.86 m, and the inside depth is measured as 88.0 cm.

(a) State Kavi's estimated diameter, then find the diameter the tape gives, to the nearest hundredth of a centimetre.

(b) Find the **percent error** of the estimated diameter, to the nearest tenth of a percent, taking the tape's value as the true one.

(c) Find the barrel's capacity in litres, to the nearest tenth, from Kavi's estimate and then from the tape's diameter.

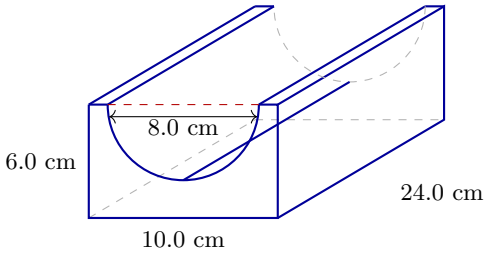
(d) Find the percent error of the estimated *capacity*, to the nearest tenth of a percent. Then show — with the two numbers, not with words alone — exactly why it is the **square** of the diameter's error factor and not twice it.

(e) Kavi needs to know (i) whether the barrel will fit through a 70 cm gateway, and (ii) how much mosquito treatment to add, at 2 mL per 10 L of water. For each, say whether the span estimate is good enough; if it is not, name an instrument that is and the precision it reads to.

Part C — Composite Solids

*In a composite solid, surfaces are **added and removed**. A face where two pieces meet belongs to neither outside surface; a face that a cut opens up belongs to it.*

6. A rectangular block of wood measures 24.0 cm long, 10.0 cm wide and 6.0 cm high. A groove whose cross section is a **semicircle** of diameter 8.0 cm is routed straight along the middle of the top face, running the whole 24.0 cm length.



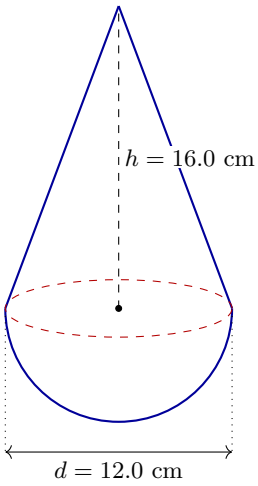
(a) Determine the volume of wood remaining, to the nearest tenth of a cubic centimetre. Write it as one subtraction, naming the base area of each prism.

(b) Determine the total surface area of the grooved block, to the nearest tenth of a square centimetre. Take every outside face, including the two ends and the floor of the groove. List what the groove *removed* and what it *added*.

(c) The block is turned into a mould: the groove is plugged at both ends and filled with wax. What is its capacity, in millilitres to the nearest tenth? ($1\ \text{cm}^3 = 1\ \text{mL}$.)

(d) A second block, also 24.0 cm long, needs a semicircular groove of capacity exactly 500 mL. What groove diameter is required, to the nearest tenth of a centimetre? Check that it still fits in a block 10.0 cm wide and 6.0 cm high.

7. A wooden spinning top is a **cone** of perpendicular height 16.0 cm joined along its base to a **hemisphere** of the same diameter, 12.0 cm. The dashed circle marks where the two pieces meet.

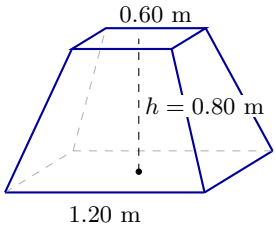


(a) Determine the total volume of the top, to the nearest tenth of a cubic centimetre, and then in litres to the nearest thousandth.

(b) Determine the total outside surface area, to the nearest tenth of a square centimetre. Say explicitly which height the cone's surface-area formula needs, find it, and state what happens to the hemisphere's flat circular base.

(c) A larger top of the same 12.0 cm diameter is to have a volume of exactly 1.50 L. What perpendicular cone height is needed, to the nearest tenth of a centimetre?

8. A concrete pier footing is the **frustum** of a square pyramid: a square base 1.20 m on a side, a square top 0.60 m on a side, and a perpendicular height of 0.80 m. The sides are flat trapezoids, and the footing sits in a hole with only the top face showing.



(a) The frustum is what is left when a smaller pyramid is cut off the top of a larger one. Using the fact that the top square is exactly half the base square, show that the complete pyramid would be 1.60 m tall, and say which scaling idea you used.

(b) Determine the volume of concrete, exactly, as (big pyramid) − (small pyramid). Then check your answer against the scaling rule: the small pyramid should be $\frac{1}{8}$ of the big one.

(c) Express that volume in litres, and in cubic feet to the nearest tenth.

(d) Ready-mix concrete costs \$185.00 per cubic metre and is sold only in whole quarter-cubic-metres. Find the exact cost of the concrete used, and the cost of what must actually be ordered.

(e) Plywood formwork lines the four sloping sides only. Find the slant height of one trapezoidal face and then the formwork area, to the nearest hundredth of a square metre.

Part D — Working Backwards

Each solid family, run in reverse. Write the formula, isolate the letter you want, then *substitute — and keep the unrounded value for any later part.*

9. Solve each for the dimension named. Round only at the end, to the precision stated.

(a) SPHERE FROM VOLUME. A spherical float holds 2.50 L. Find its diameter, to the nearest tenth of a centimetre.

(b) CONE FROM LATERAL AREA. A cone's curved surface is 0.25 m^2 and its base radius is 12.0 cm. Find the slant height, then the perpendicular height, each to the nearest tenth of a centimetre.

(c) CYLINDER FROM SURFACE AREA. A closed cylinder has total surface area 600 cm^2 and radius 5.0 cm. Find its height, to the nearest tenth of a centimetre.

(d) PRISM FROM CAPACITY. A rectangular tank holds 450 L. Its base is 1.20 m by 0.75 m. Find its depth, to the nearest centimetre.

10. A closed cylindrical can has a total surface area of 400 cm^2 and a height of 10.0 cm. This time the *radius* is unknown, and $SA = 2\pi r^2 + 2\pi rh$ is a quadratic in r. Write it in the form $ar^2 + br + c = 0$, solve it, and reject the root that cannot be a radius. Give r to the nearest tenth of a centimetre, and then the can's capacity in millilitres to the nearest millilitre.

11. A hemispherical grain dome is built **with a concrete floor**, and the total area of concrete and shell together is 120 m^2.

(a) State which hemisphere surface-area formula that wording selects, and why. Then find the radius, to the nearest hundredth of a metre.

(b) Find the dome's capacity in litres, to the nearest litre, carrying the unrounded radius.

(c) Had the wording been "the shell has area 120 m^2", what radius would follow? State the ratio of the two radii as an exact surd, and explain why it is not $\frac{2}{3}$.

Part E — Reasoning and Error Analysis

Every claim below is false or incomplete. Say precisely what is wrong, ***and put a number on it.***

12. **"Doubling the radius doubles the volume."** A cylindrical tank has radius 1.2 m and height 3.0 m.

(a) Find the volume as it stands, to the nearest hundredth of a cubic metre. Then find it with the radius doubled and the height unchanged, and state the exact factor.

(b) Now double *every* dimension. State the exact factor, and say which exponent in $V = \pi r^2 h$ each factor comes from.

(c) Repeat for the surface area. Doubling all dimensions multiplies SA by 4 — but doubling the radius alone does *not*. Find that factor exactly, and explain in one sentence why the tidy k^2 rule does not apply.

13. **"Solid A has surface area** 162 **cm^2 and solid B has volume** 64 **cm^3, so A is bigger."** Solid A is a rectangular prism 1.0 cm by 1.0 cm by 40.0 cm; solid B is a cube of edge 4.0 cm.

(a) Find *both* the surface area and the volume of *both* solids, and set them out in a small table. State the single sentence that makes the original claim meaningless.

(b) Which solid holds more water? Which needs more wrapping paper? Which loses heat faster? Answer each with the number that decides it, and give each solid's surface-area-to-volume ratio with its correct unit.

14. **"**3.14 **is close enough."** A spherical gas holder has a radius of 9.00 m.

(a) Find the volume with exact π and again with 3.14, each to the nearest hundredth of a cubic metre. Give the difference in cubic metres and in litres.

(b) Find the surface area both ways. A protective coating costs \$62.00 per square metre: how much would the 3.14 quotation be out by?

(c) Give the percent error of 3.14 itself, to four significant figures. Show that the percent error in the volume is the *same*, and explain why this error — unlike a scaling factor — is not cubed.

15. **Two errors in four lines.** Asked for the surface area of a closed cylinder of radius 40 cm and height 1.2 m, **in square metres**, a student wrote

$$SA = 2\pi(40)^2 + 2\pi(40)(1.2) = 3296\pi = 10\,354.7 \text{ cm}^2, \qquad 10\,354.7 \div 100 = 103.5 \text{ m}^2.$$

(a) Name both errors exactly, and say which one a unit-factor layout would have caught on sight.

(b) Give the correct surface area to the nearest hundredth of a square metre, working entirely in metres.

(c) Redo it entirely in centimetres and convert at the end, and show the two routes agree. State the factor by which the student's answer is wrong.

16. **Two heights, and one flat circle.** Find the mistake in each report and correct it.

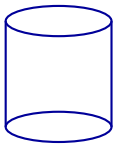
(a) A cone has base radius 9.0 cm and slant height 15.0 cm. A student writes $V = \frac{1}{3}\pi(9.0)^2(15.0) = 1272.3 \text{ cm}^3$. Give the correct volume to the nearest tenth, and the exact factor by which the student is out.

(b) A solid glass hemisphere of radius 5.0 cm rests on a felt pad. Asked for "the glass surface you can touch", a student answers $3\pi r^2 = 235.6 \text{ cm}^2$. Give the correct area, and write the one phrase in the question that decides it.

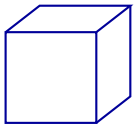
(c) A hemispherical dome tent of radius 1.8 m is sold "complete with sewn-in groundsheet". Give the total area of fabric to the nearest hundredth of a square metre, and the area if the groundsheet were sold separately.

Part F — Challenge

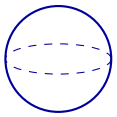
17. **Least material for the same capacity.** Three closed containers must each hold exactly 2000 cm^3: **A** a cylinder whose height equals its diameter, **B** a cube, **C** a sphere.



A cylinder, $h = 2r$



B cube



C sphere

(a) Find each container's dimensions, to the nearest hundredth of a centimetre. Each part is a formula run backwards — write the rearrangement before you substitute.

(b) Find each surface area, to the nearest hundredth of a square centimetre, and rank them from least material to most.

(c) Sheet material costs \$0.0042 per square centimetre. Give the three costs to the nearest cent, and the percentage saved by the best over the worst, to the nearest tenth.

(d) The winner is not the container anyone sells. Give two practical reasons why, and name one real product that *is* close to the winning shape and one everyday object close to the runner-up.

18. **When the surface area and the volume are the same number.**

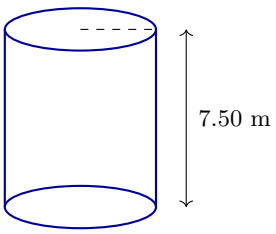
(a) Find the radius for which a sphere's surface area and volume are numerically equal. Do it algebraically, not by trial. Then do the same for a cube's edge, and for the radius of a cylinder with $h = 2r$.

(b) For the sphere you found, give SA and V exactly in terms of π and confirm the two numbers agree. Then rewrite *the same sphere* in millimetres and show that the numbers no longer agree — give both, and the factor between them.

(c) Explain, using the k^2 and k^3 rules, why changing the unit must break the equality, and why *every* sphere therefore has exactly one unit in which SA and V are numerically equal.

(d) A sphere has radius 5.0 cm. Find the length of the unit — in centimetres — in which this sphere's surface area and volume would be numerically equal. Then state, in one sentence, what "$SA = V$" actually tells you about a solid, and what it does not.

19. **A scale model.** A cylindrical water tower has a diameter of 6.40 m and a height of 7.50 m. A display model is built at a scale of 1 : 25.



(a) Determine the real tower's capacity, in cubic metres to the nearest thousandth and in litres to the nearest litre.

(b) State the model's diameter and height in centimetres. Then give the model's capacity **two ways**: by dividing the real capacity by the correct power of 25, and by substituting the model's own dimensions. Give it in litres to the nearest thousandth and in millilitres in scientific notation to three significant figures.

(c) The real tower's curved wall and flat top are painted (not the base). Find that area to the nearest thousandth of a square metre, then the model's, in square metres and in square centimetres. Check it both ways as in (b).

(d) A classmate says "the model is $\frac{1}{25}$ the size, so it holds $\frac{1}{25}$ as much." Give the litres that claim produces, and the exact factor by which it is wrong.

(e) Write the sentence that links the scale factors k, k^2, k^3 to the unit factors 100, 100^2, 100^3 used in converting m, m^2 and m^3 to cm, cm^2 and cm^3. Verify it on your answer to (c).

Part G — Estimating and Converting in the Field

The last two questions put the instrument down and pick the referent up, then put the referent down and pick the imperial tables up.

20. **A referent, used three times over.** Priya estimates the inside of a shipping container without a tape. She walks its length in 7 paces and its width in 3 paces, taking her pace to be 0.90 m; she measures its height as 3 arm-spans, taking her span to be 0.80 m. Her counts are exact. Her pace is truly 0.86 m and her span is truly 0.76 m.

(a) State Priya's three estimated dimensions and the three true dimensions, and the percent error of each, to the nearest hundredth of a percent.

(b) Find the estimated volume and the true volume, each to the nearest thousandth of a cubic metre, and the percent error of the volume, to the nearest hundredth of a percent.

(c) Show that the volume's error factor is the **product** of the three linear error factors, not their sum. Give both numbers, and say which of k, k^2, k^3 is at work here and why it is not a clean k^3.

(d) Give the container's true capacity in litres, in scientific notation to five significant figures, and state why litres is a defensible unit here and cubic centimetres is not.

(e) Priya needs to know (i) roughly how many pallets will fit, and (ii) whether a beam 6.00 m long will go in. For each, say whether the paced estimate is fit for purpose; where it is not, name the instrument you would use and the precision it reads to.